

BFS Traversal

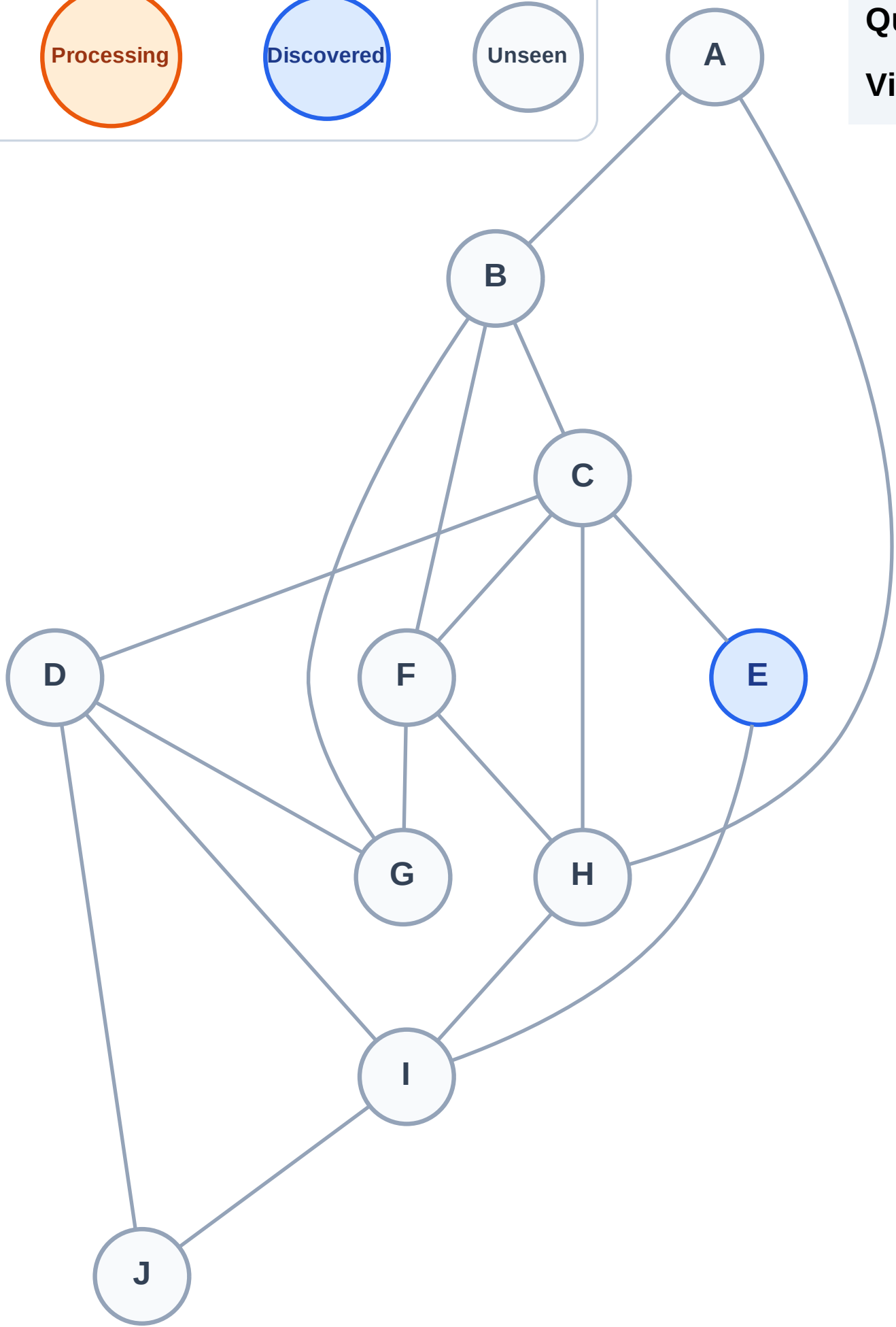
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

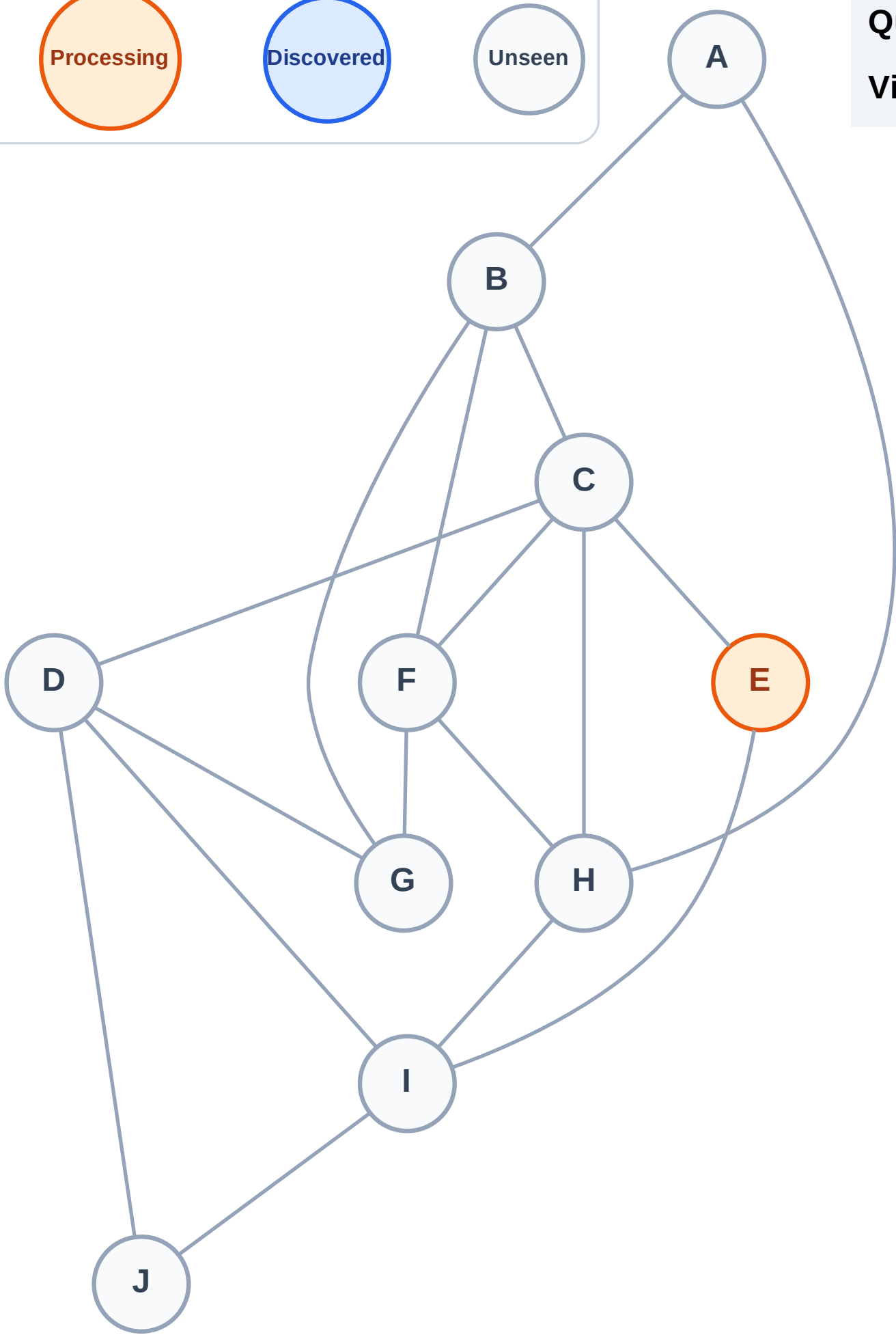
Step 2: Process node E

Legend



Queue: (empty)

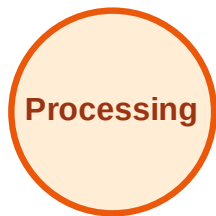
Visited: (none)



BFS Traversal

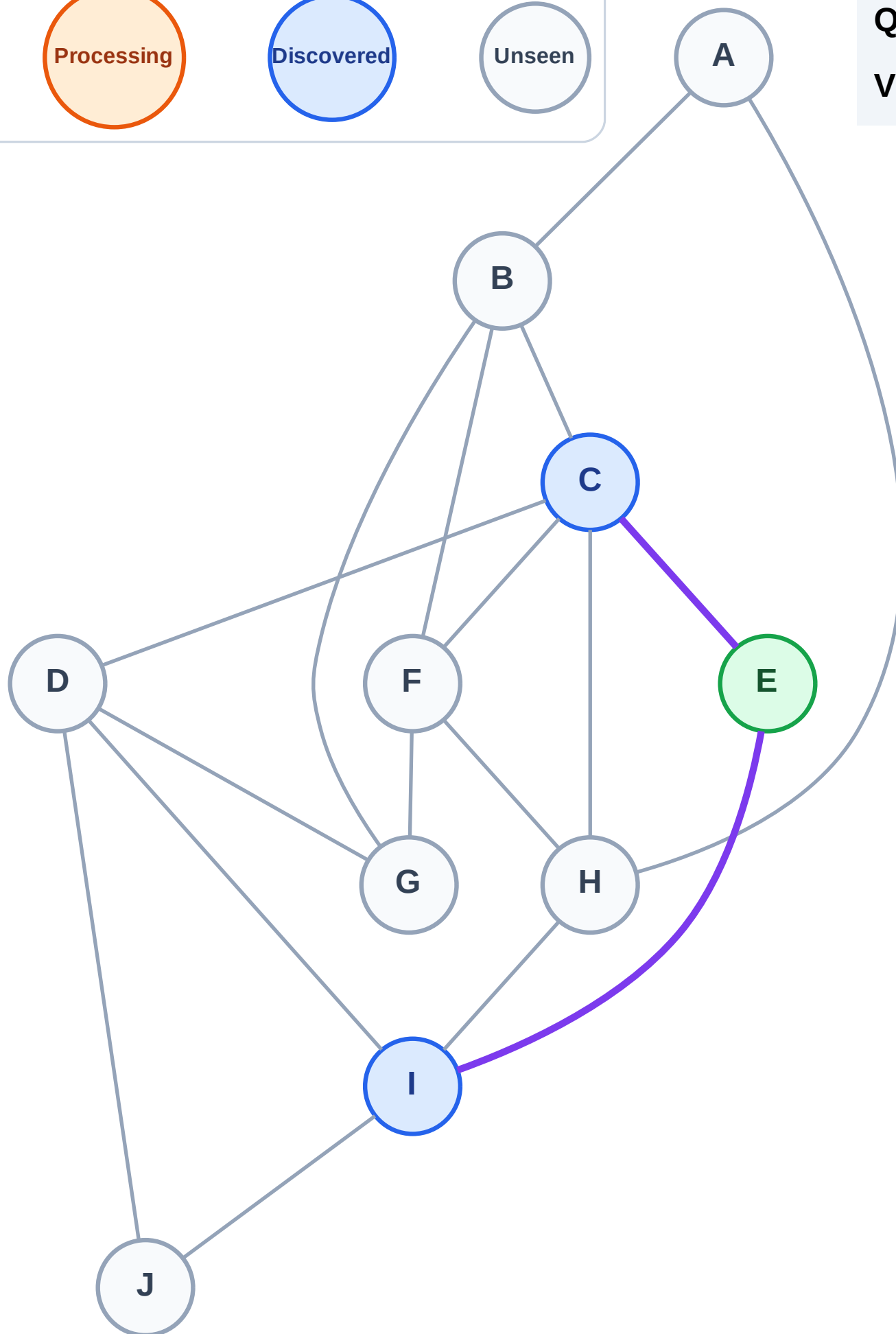
Step 3: Discover C, I from E

Legend



Queue: C → I

Visited: E



BFS Traversal

Step 4: Process node C

Legend

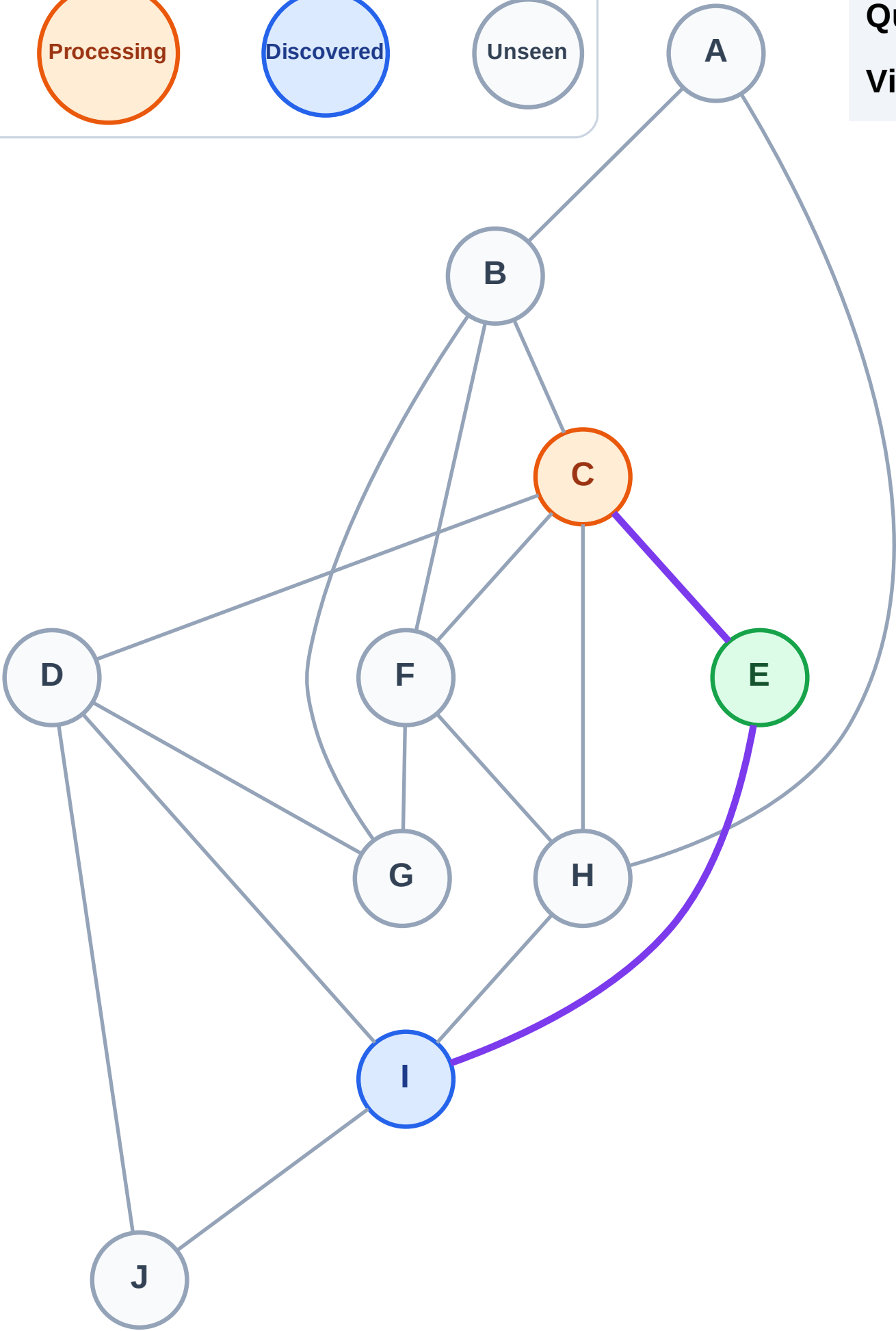
Complete

Processing

Discovered

Unseen

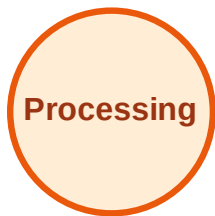
Queue: I
Visited: E



BFS Traversal

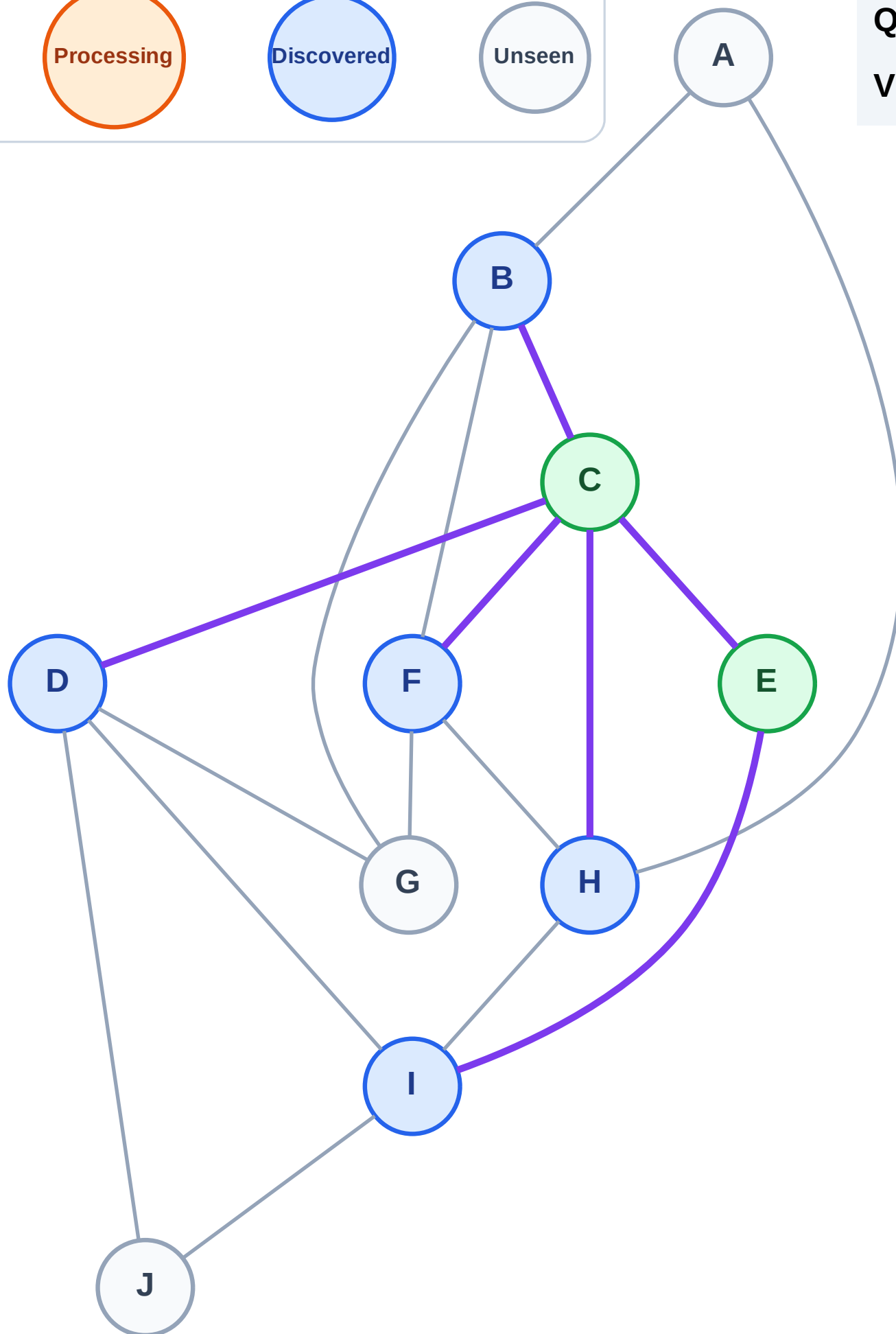
Step 5: Discover B, D, F, H from C

Legend



Queue: I → B → D → F → H

Visited: C, E



BFS Traversal

Step 6: Process node I

Legend

Complete

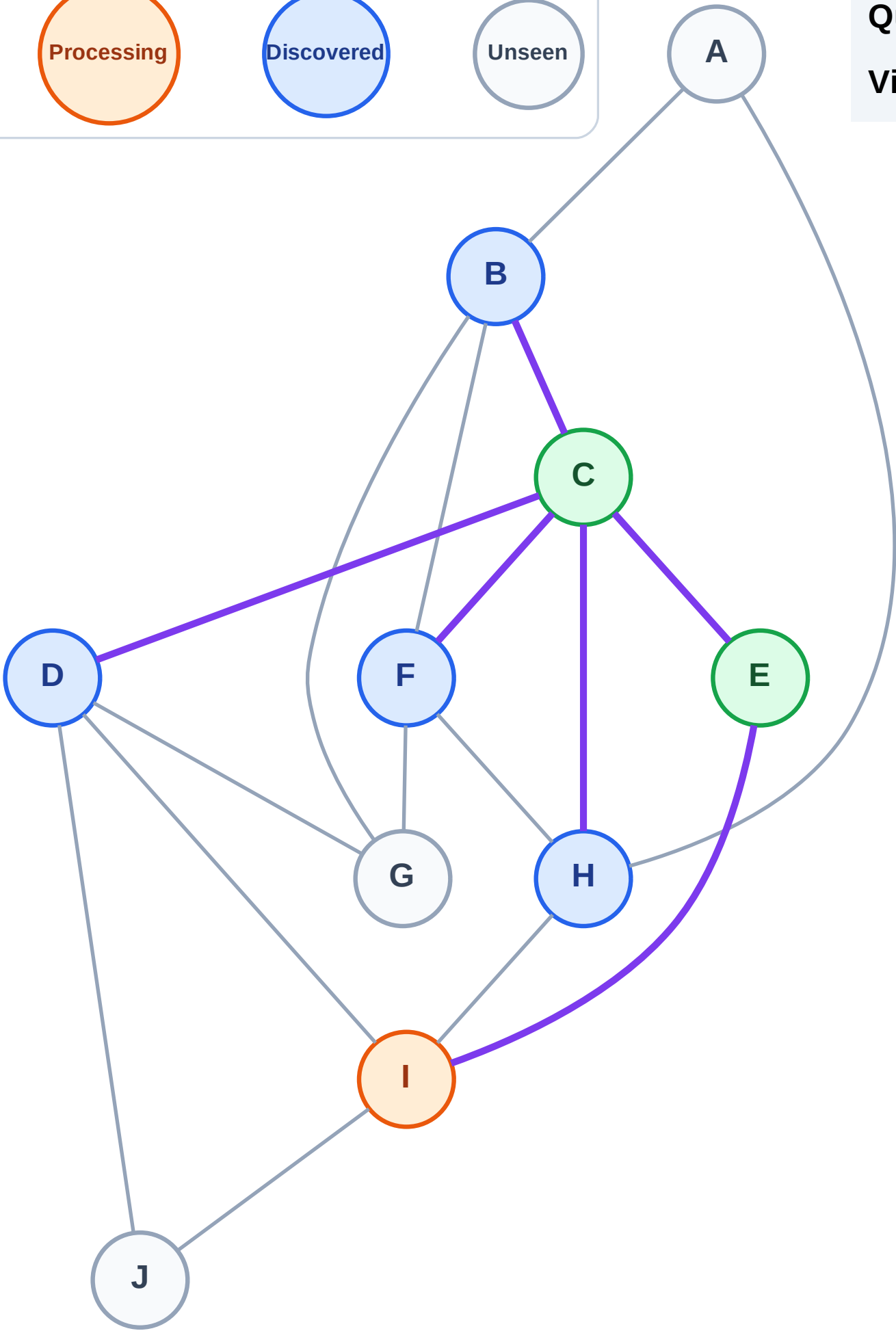
Processing

Discovered

Unseen

Queue: B → D → F → H

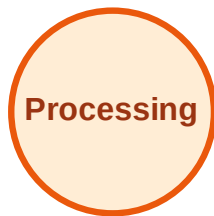
Visited: C, E



BFS Traversal

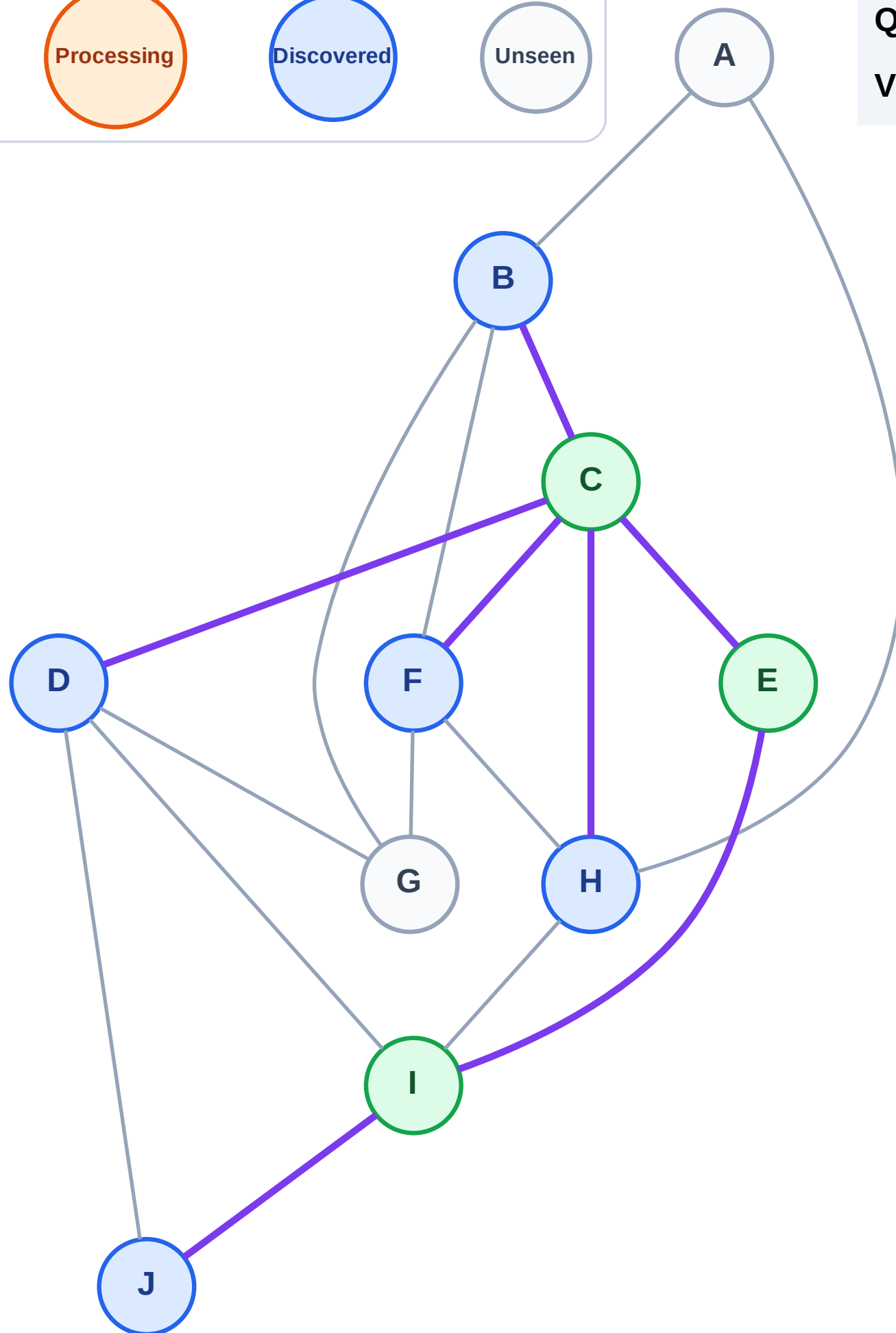
Step 7: Discover J from I

Legend



Queue: B → D → F → H → J

Visited: C, E, I



BFS Traversal

Step 8: Process node B

Legend

Complete

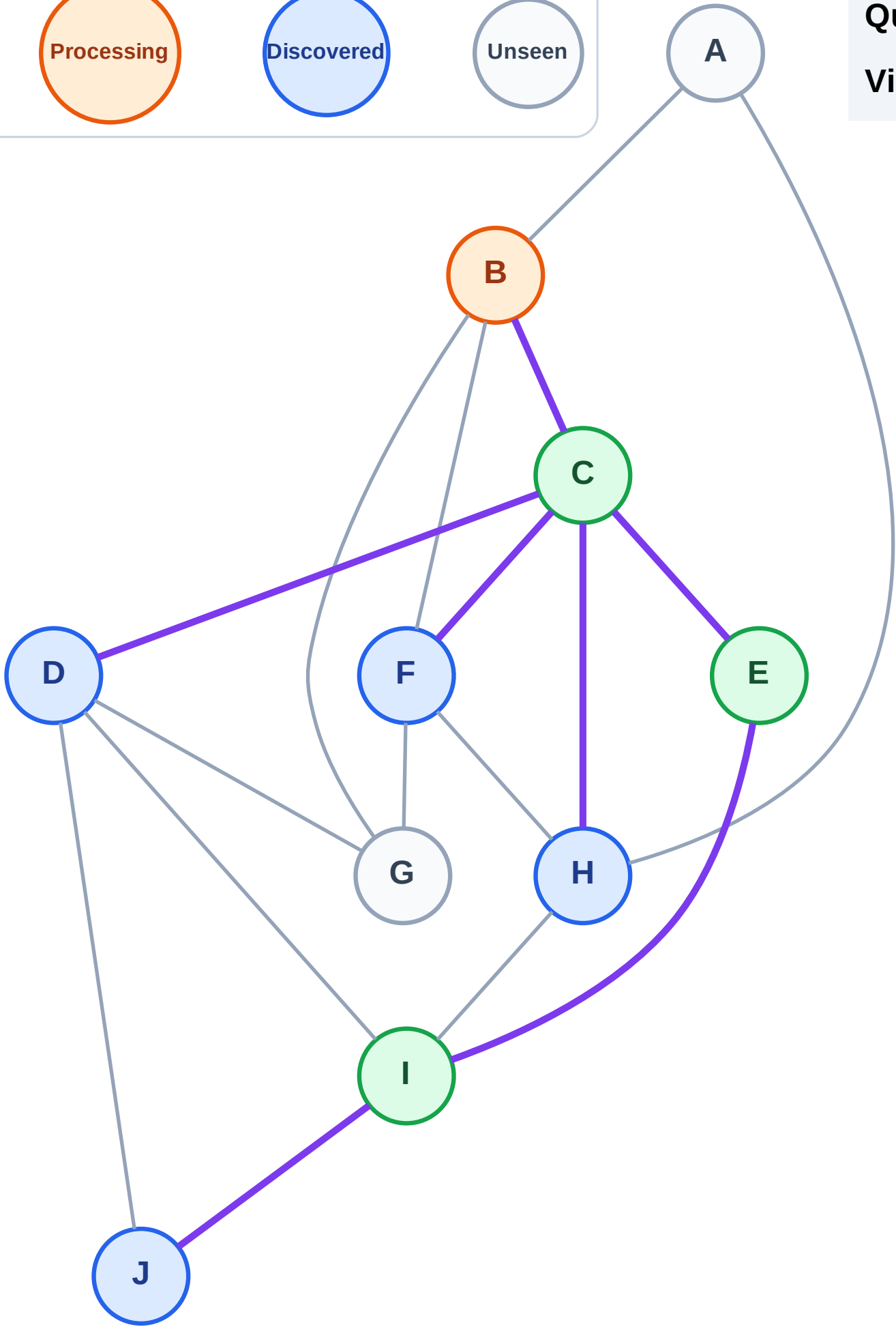
Processing

Discovered

Unseen

Queue: D → F → H → J

Visited: C, E, I



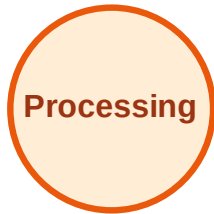
BFS Traversal

Step 9: Discover A, G from B

Legend



Complete



Processing

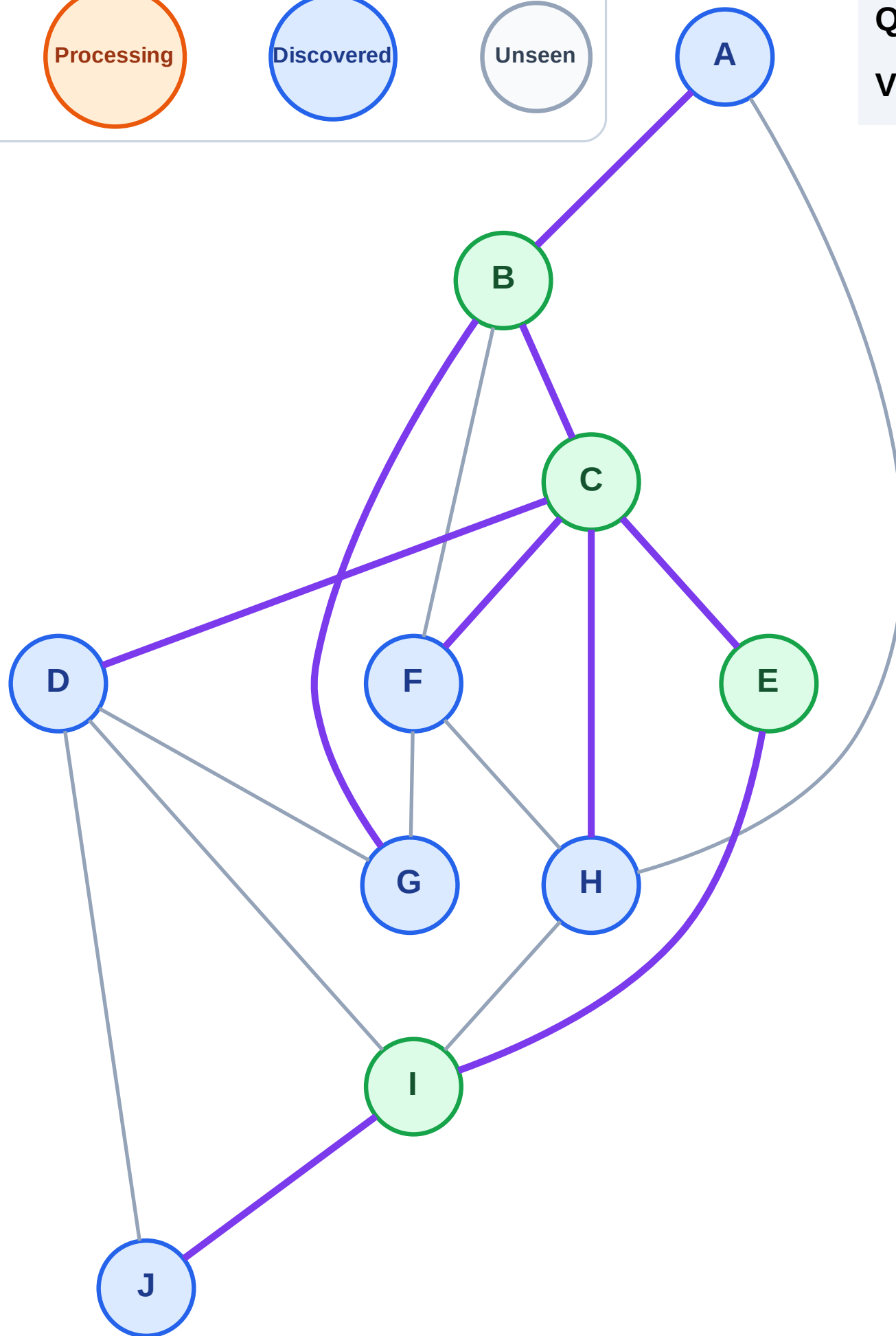


Discovered



Unseen

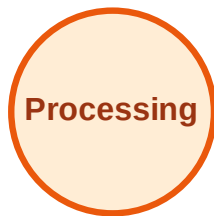
Queue: D → F → H → J → A → G
Visited: B, C, E, I



BFS Traversal

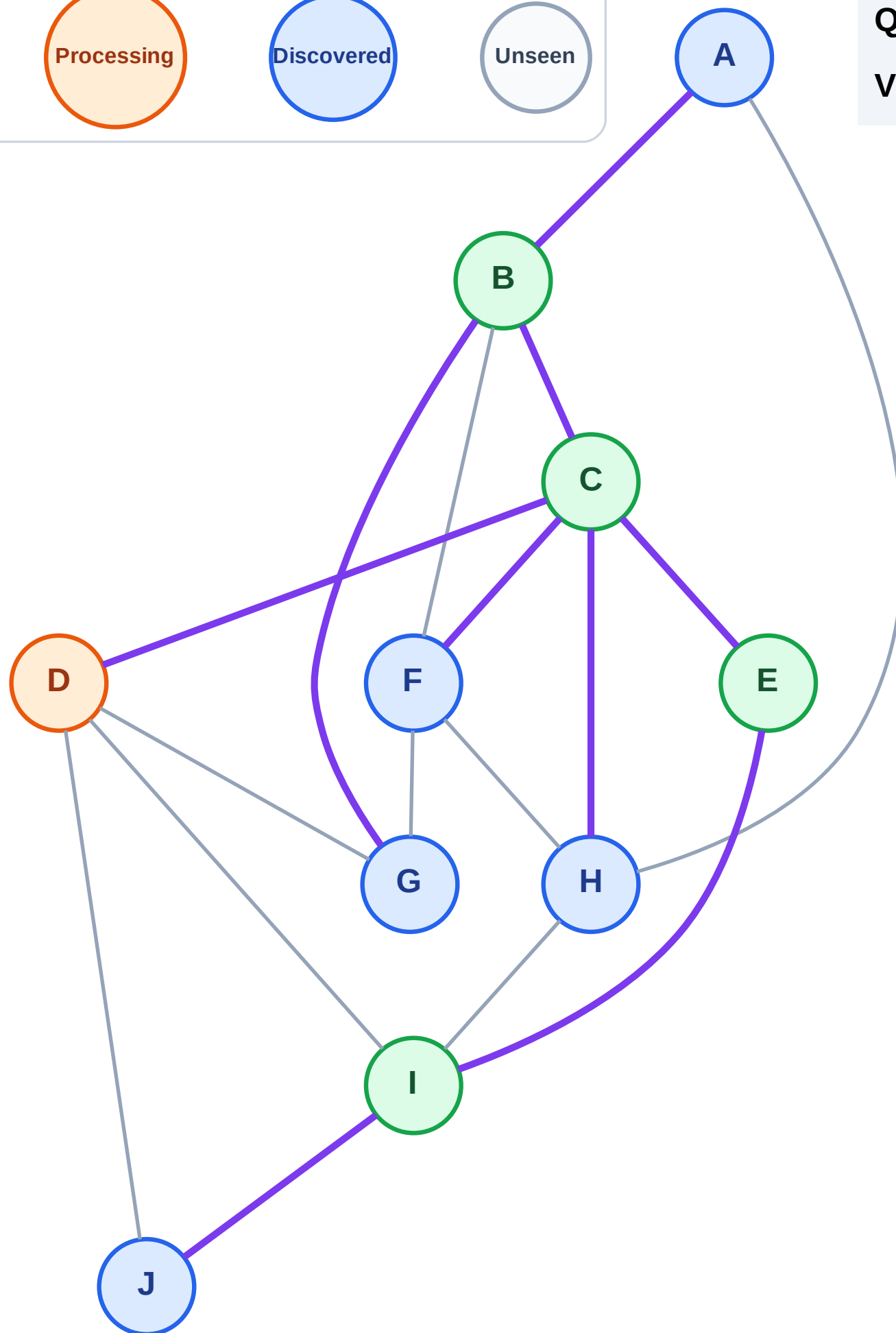
Step 10: Process node D

Legend



Queue: F → H → J → A → G

Visited: B, C, E, I



BFS Traversal

Step 11: No new neighbors from D

Legend

Complete

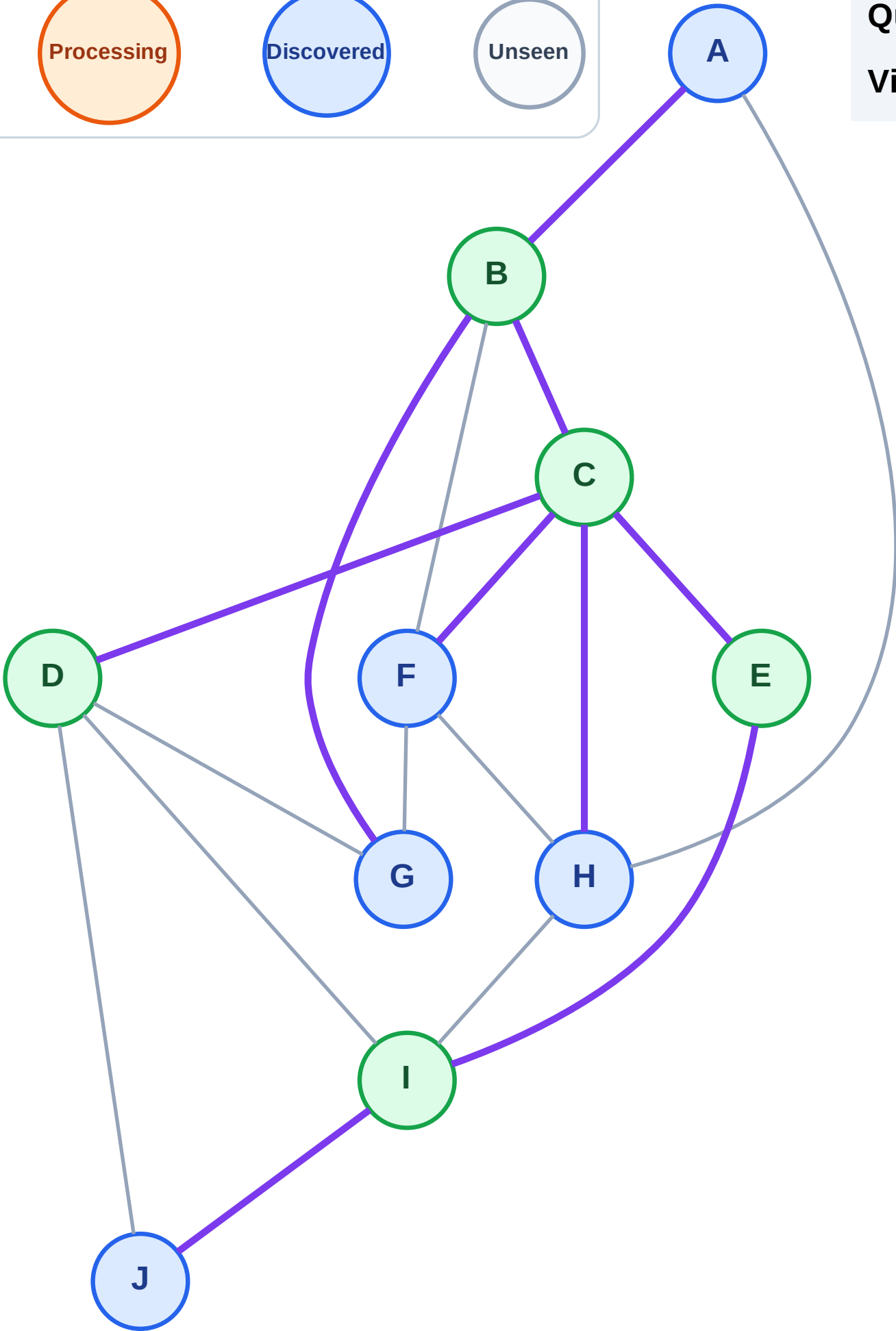
Processing

Discovered

Unseen

Queue: F → H → J → A → G

Visited: B, C, D, E, I



BFS Traversal

Step 12: Process node F

Legend

Complete

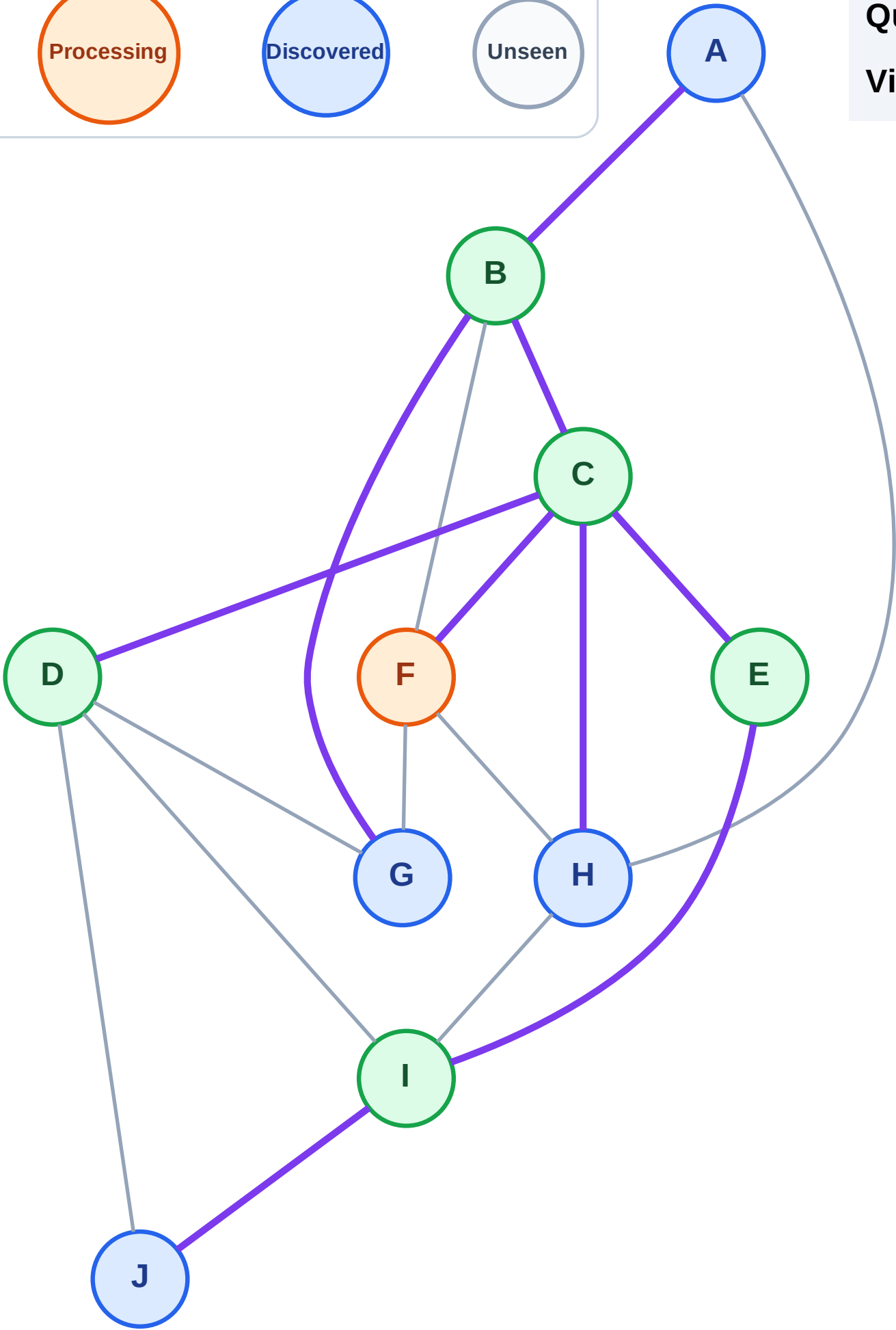
Processing

Discovered

Unseen

Queue: H → J → A → G

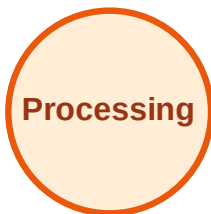
Visited: B, C, D, E, I



BFS Traversal

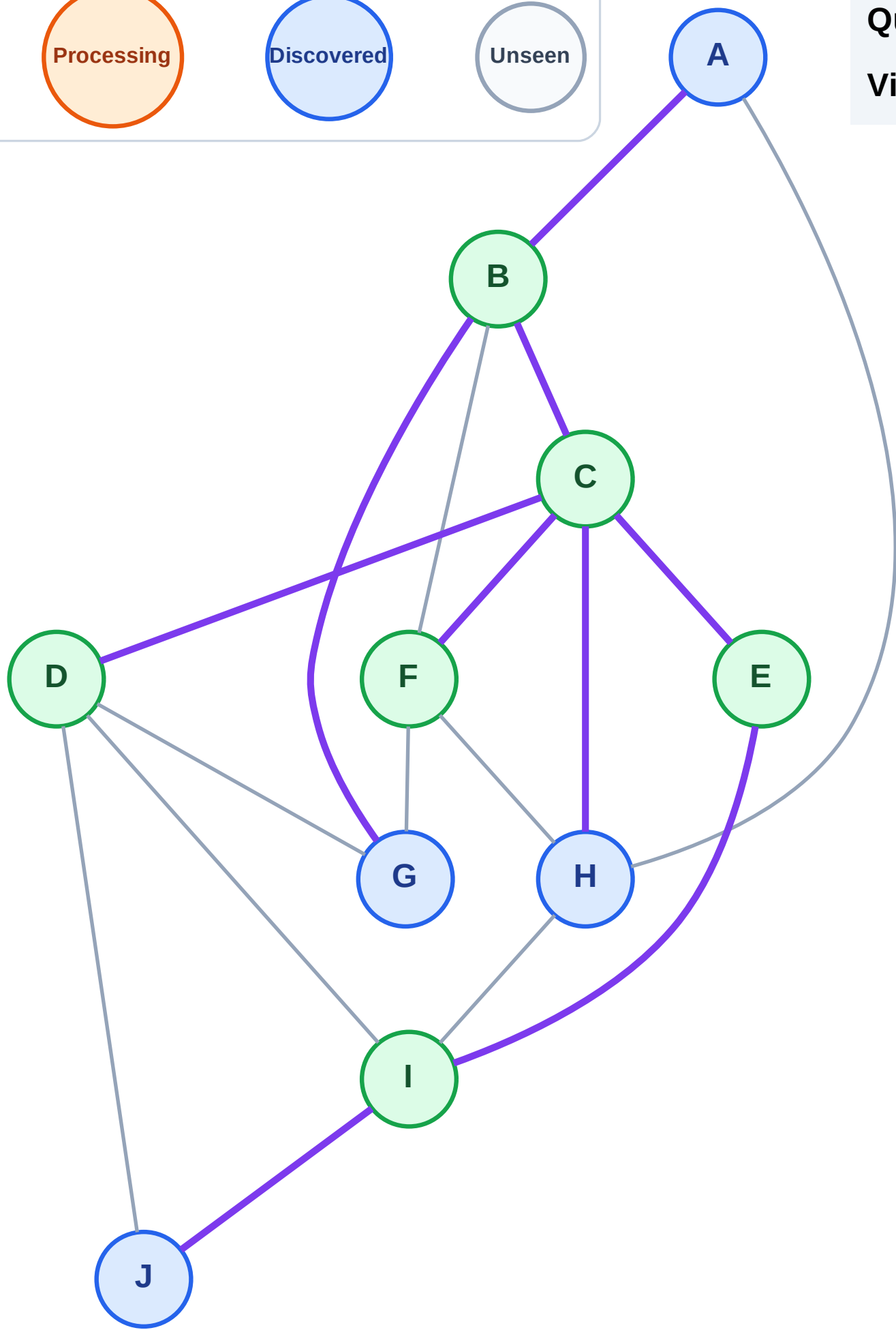
Step 13: No new neighbors from F

Legend



Queue: H → J → A → G

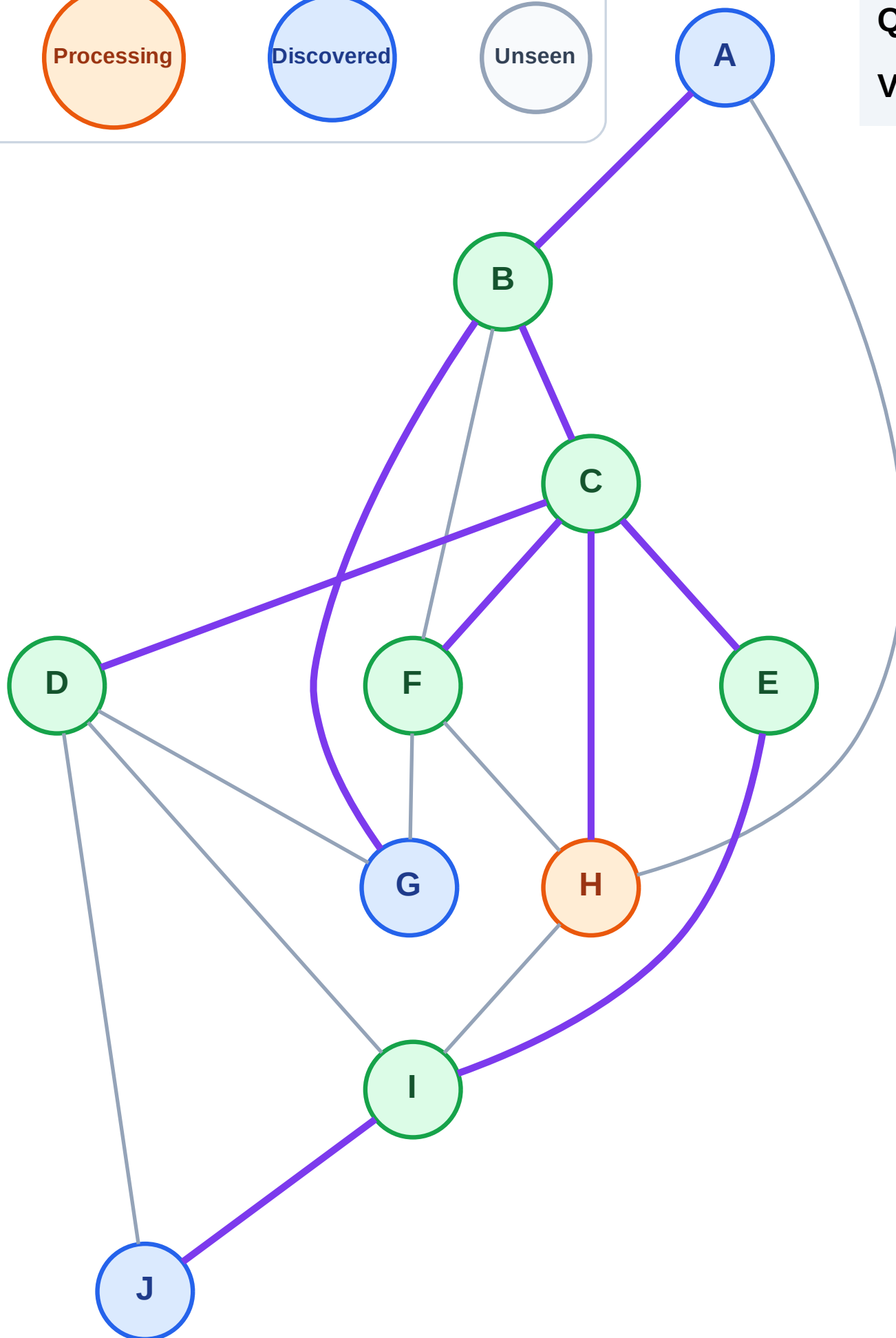
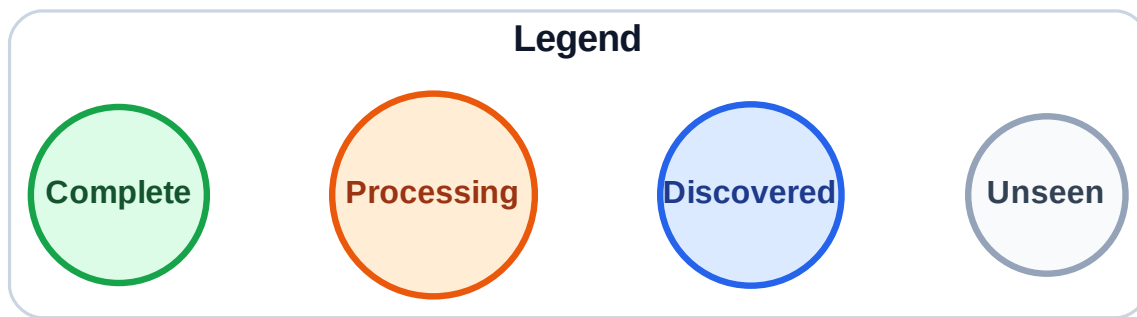
Visited: B, C, D, E, F, I



BFS Traversal

Step 14: Process node H

Queue: J → A → G
Visited: B, C, D, E, F, I



BFS Traversal

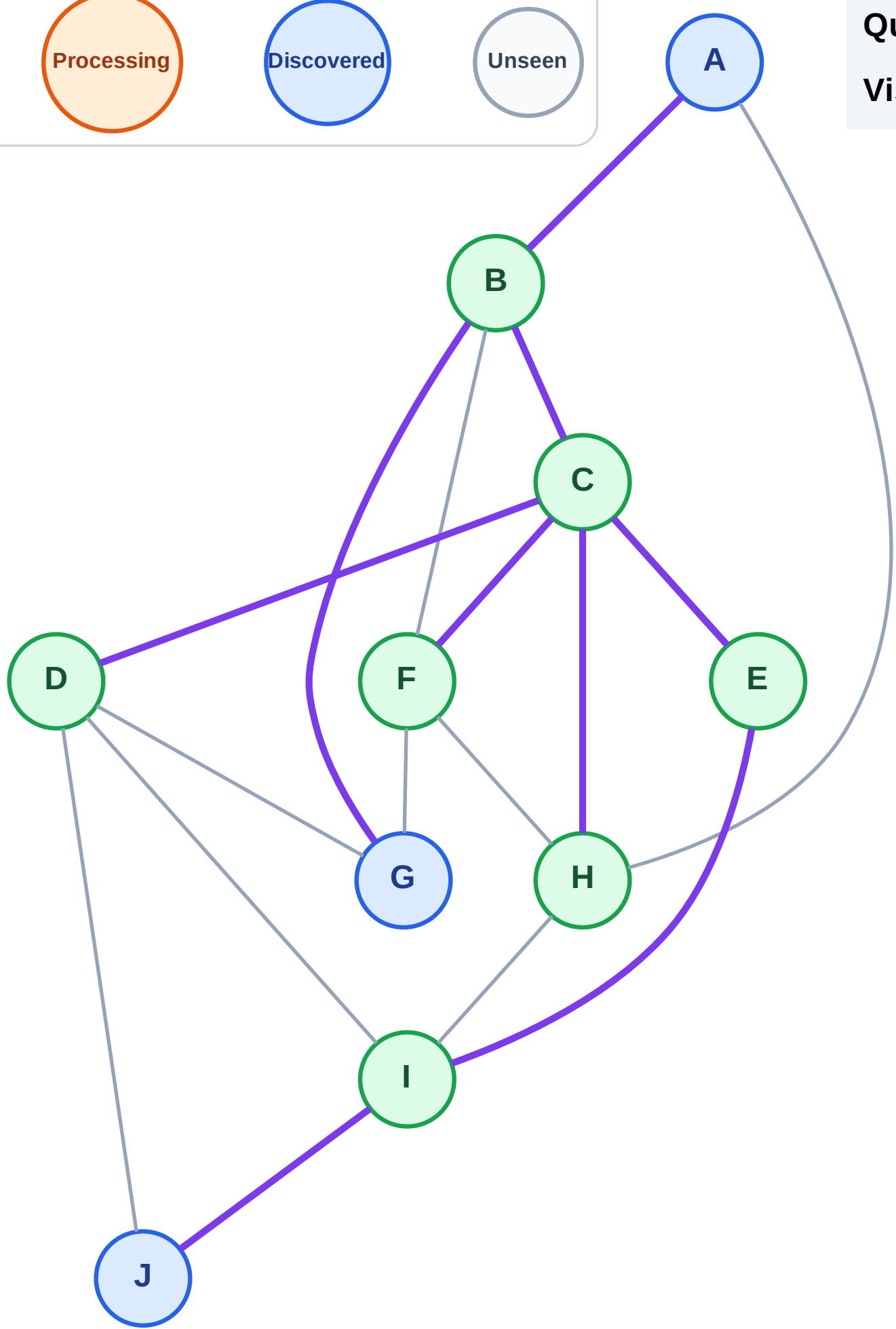
Step 15: No new neighbors from H

Legend



Queue: J → A → G

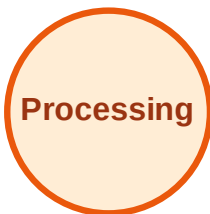
Visited: B, C, D, E, F, H, I



BFS Traversal

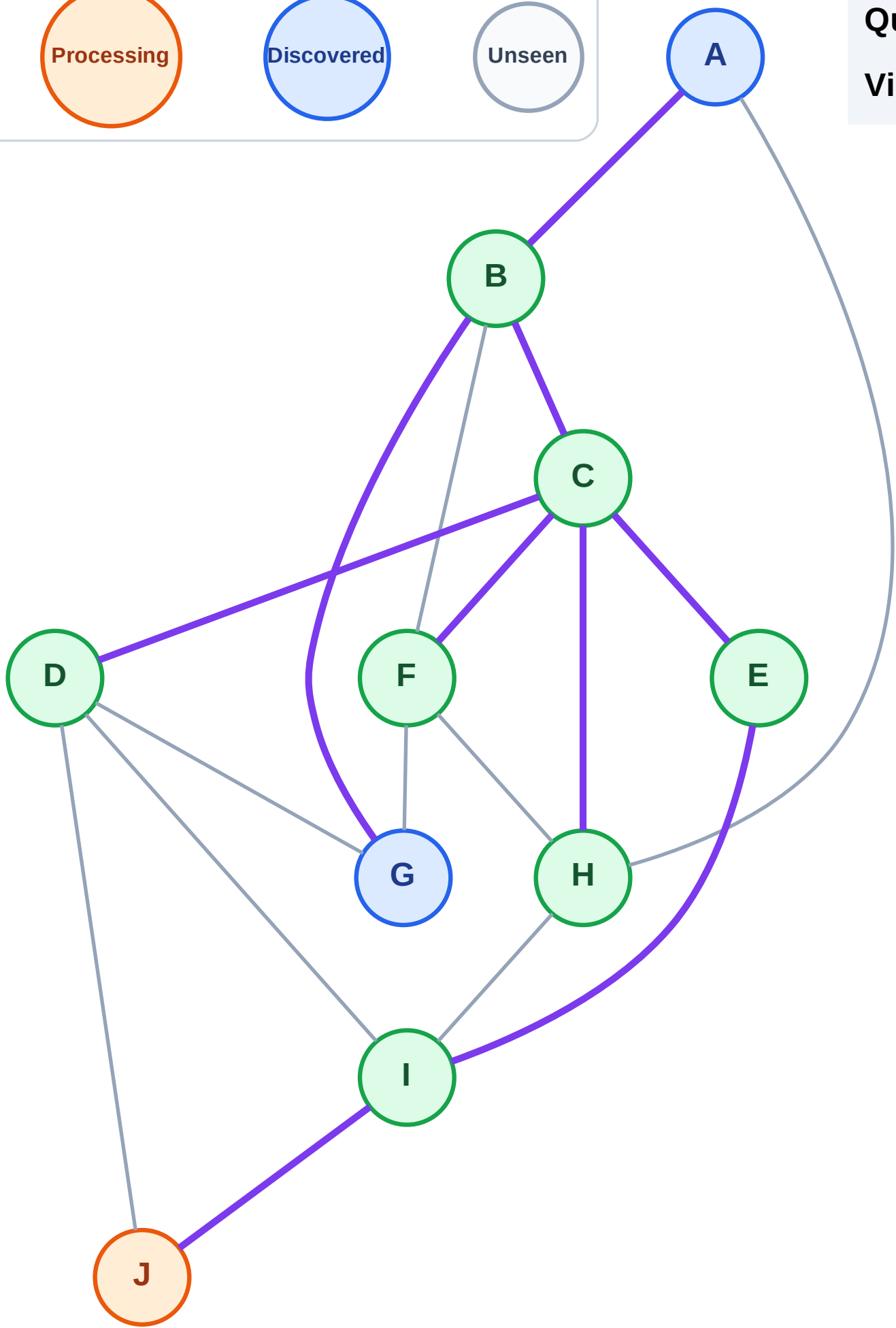
Step 16: Process node J

Legend



Queue: A → G

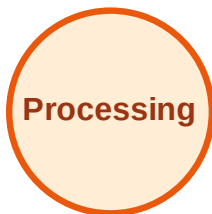
Visited: B, C, D, E, F, H, I



BFS Traversal

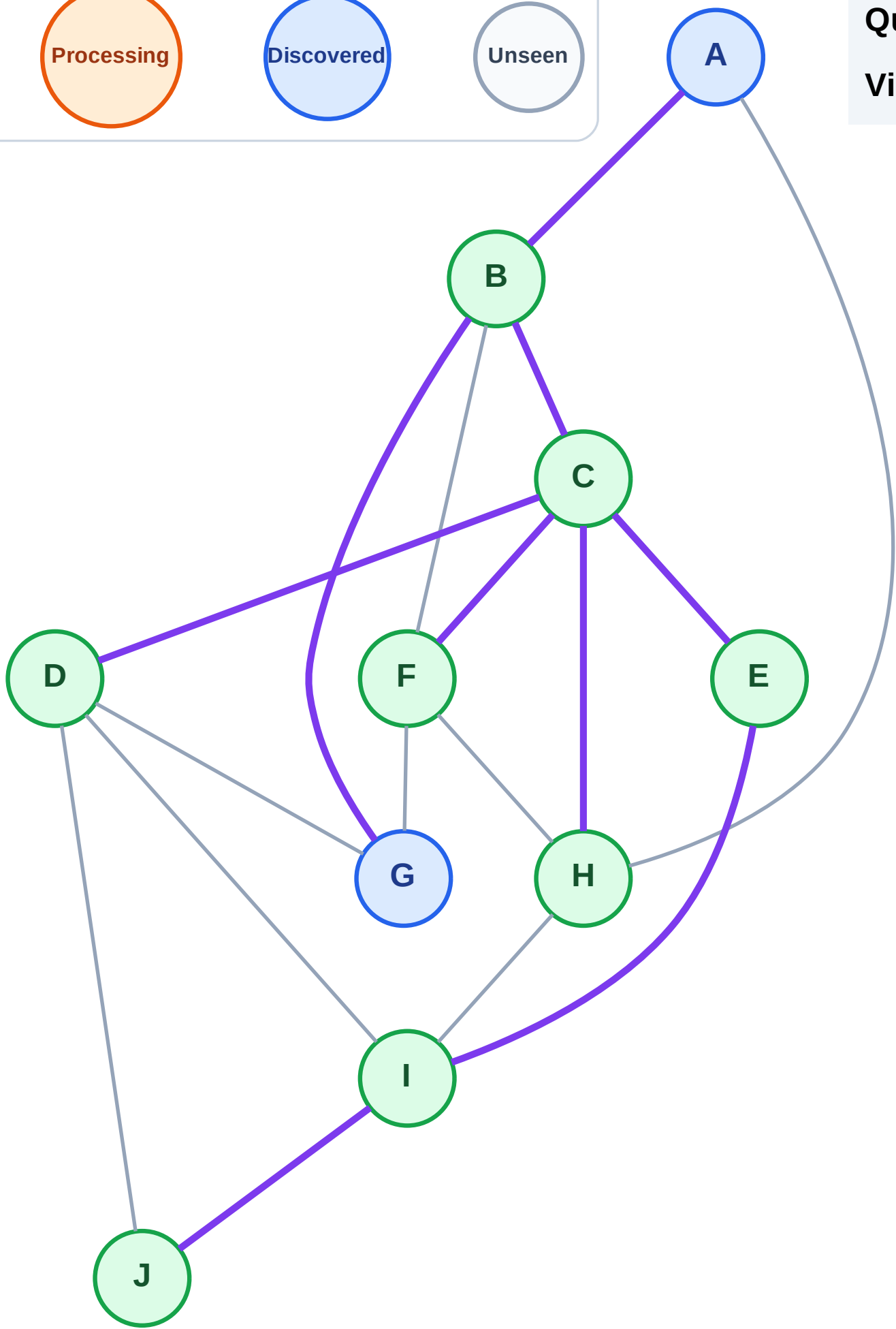
Step 17: No new neighbors from J

Legend



Queue: A → G

Visited: B, C, D, E, F, H, I, J



BFS Traversal

Step 18: Process node A

Legend

Complete

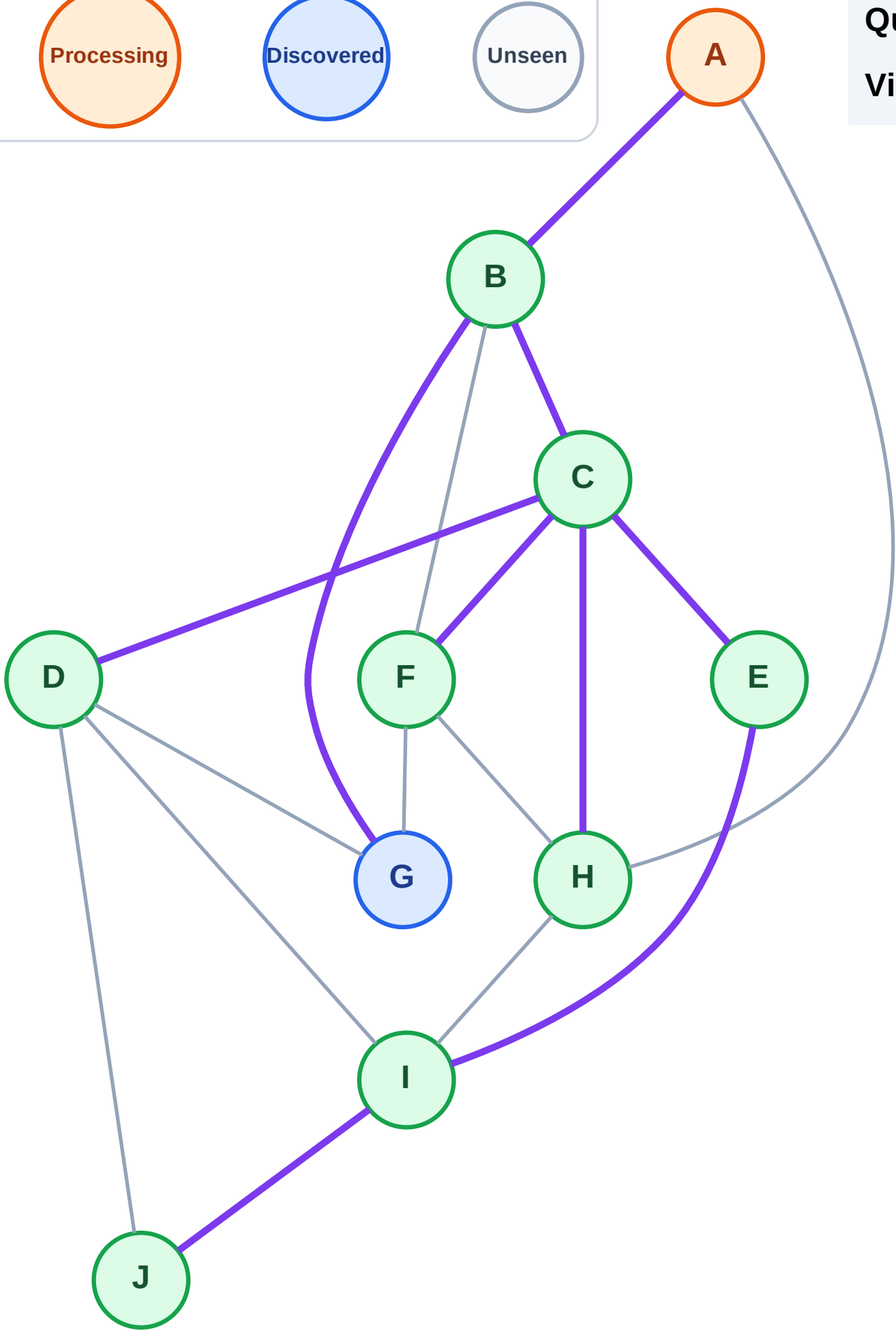
Processing

Discovered

Unseen

Queue: G

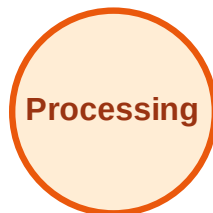
Visited: B, C, D, E, F, H, I, J



BFS Traversal

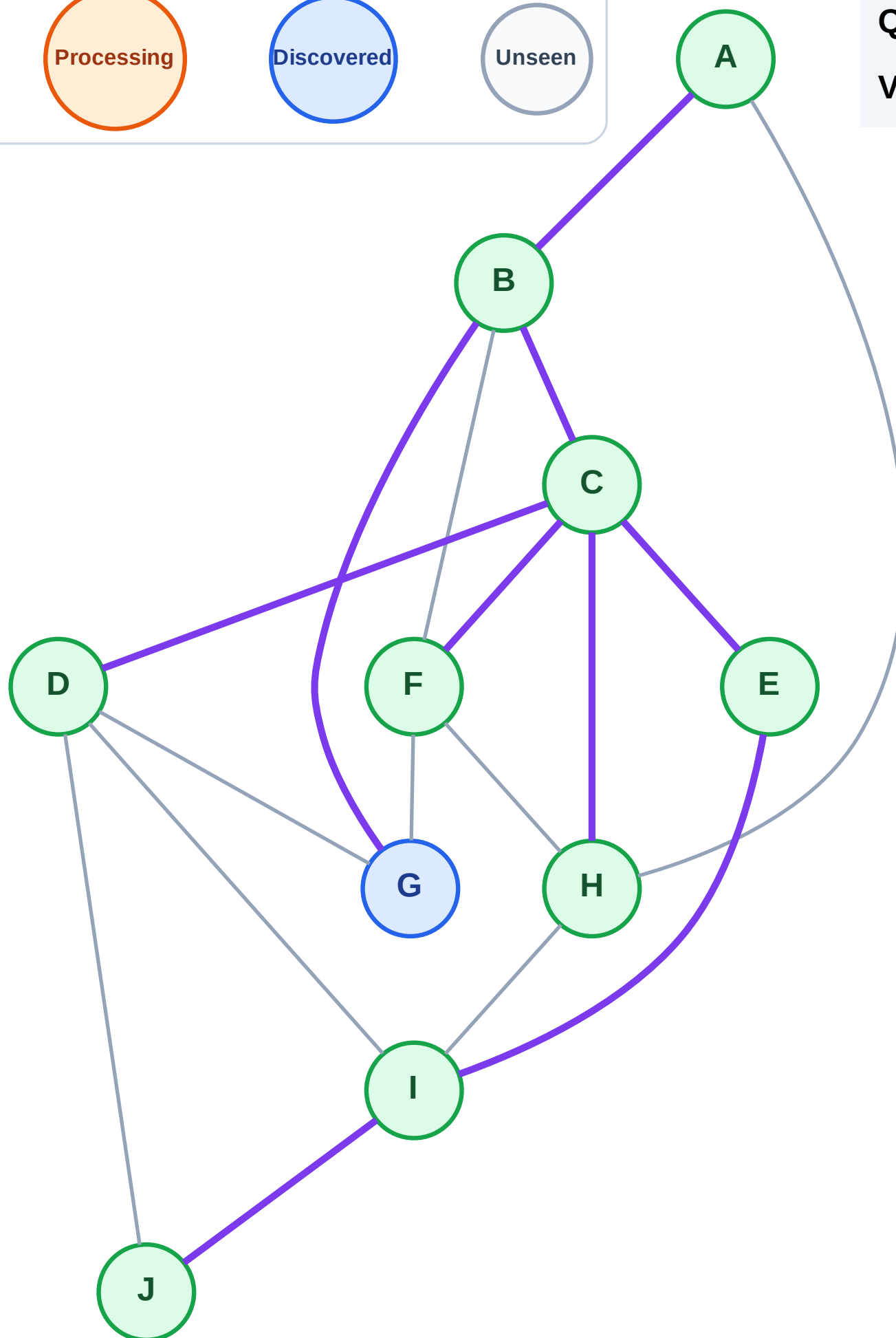
Step 19: No new neighbors from A

Legend



Queue: G

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

Step 20: Process node G

Legend

Complete

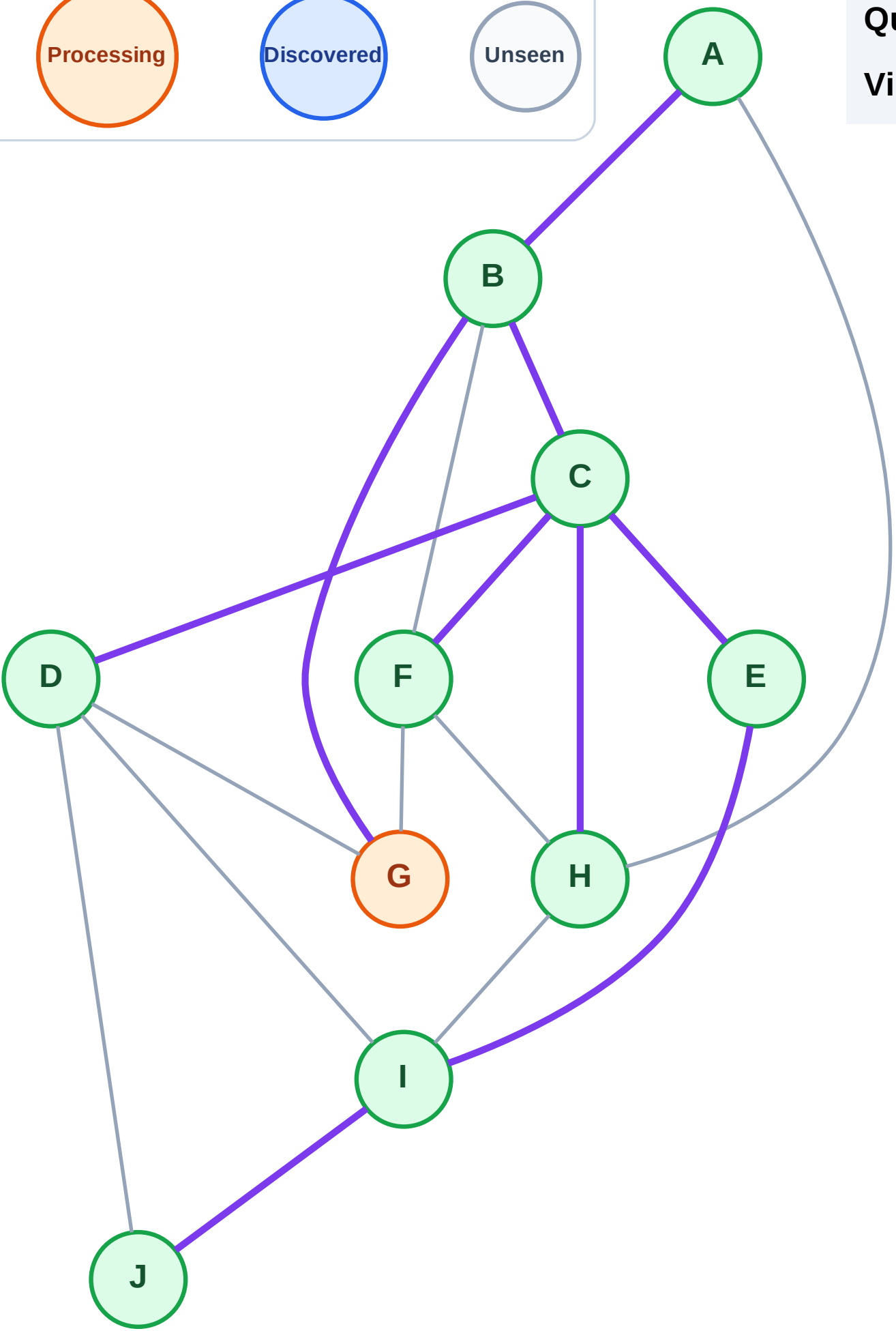
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

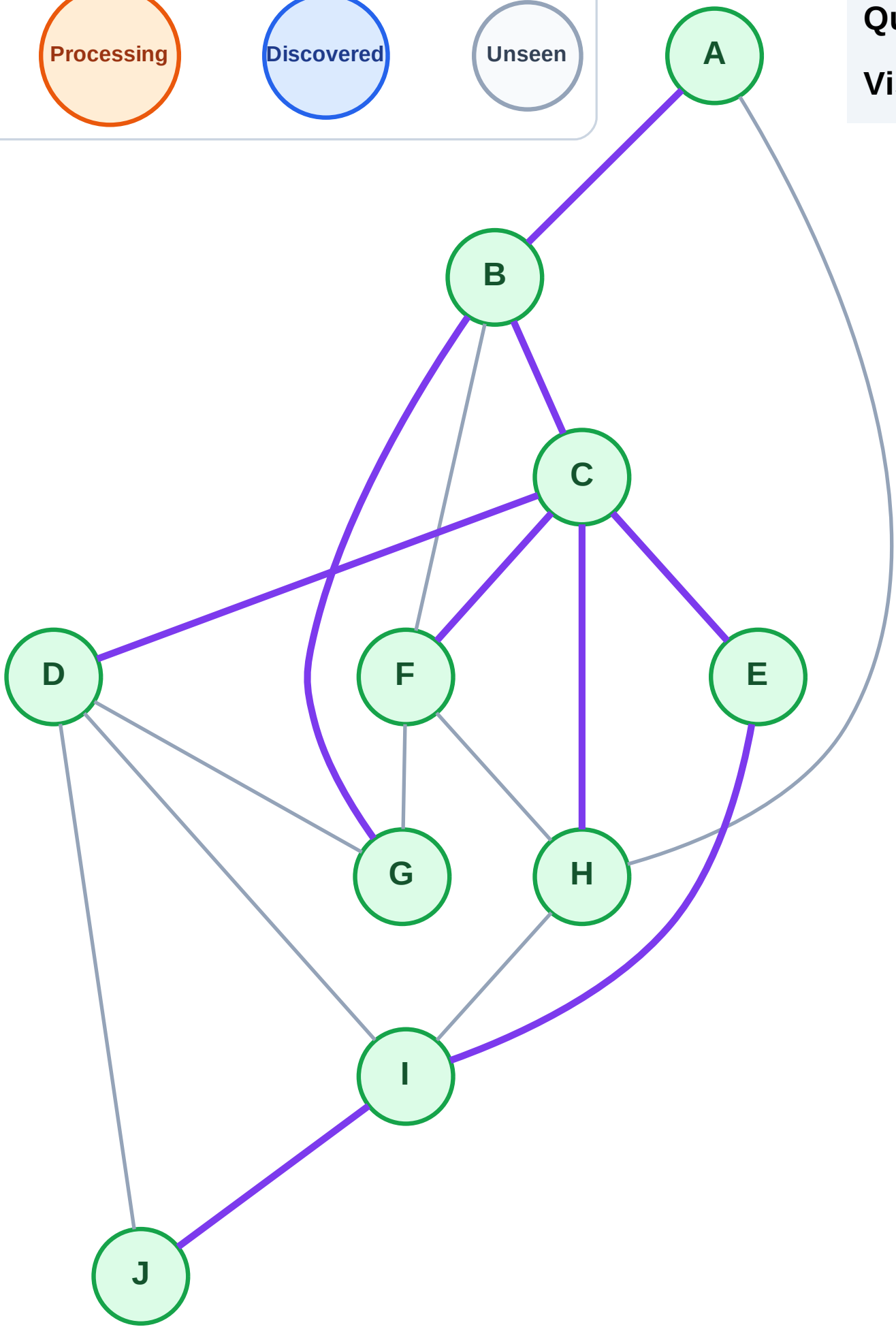
Step 21: No new neighbors from G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

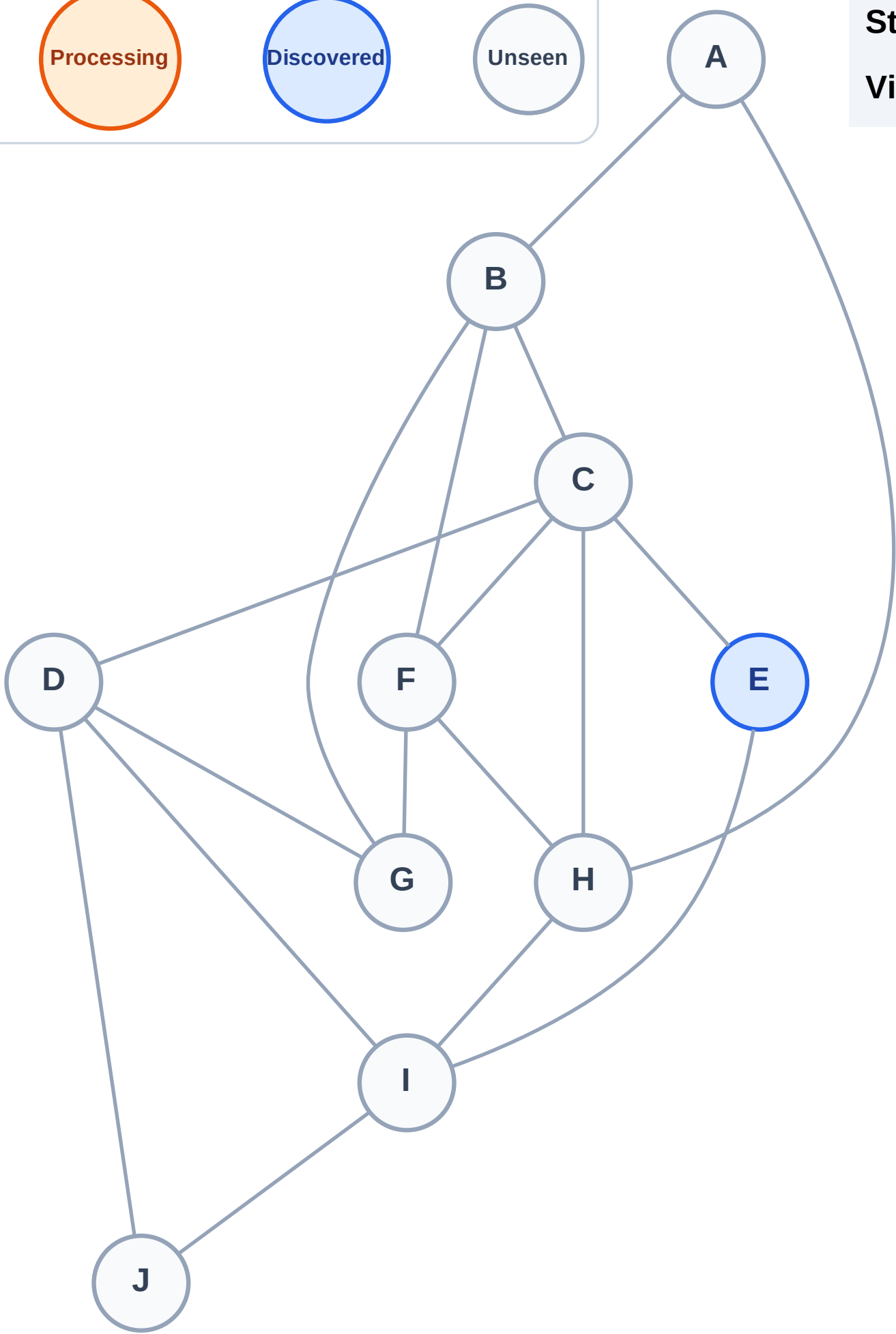
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

Complete

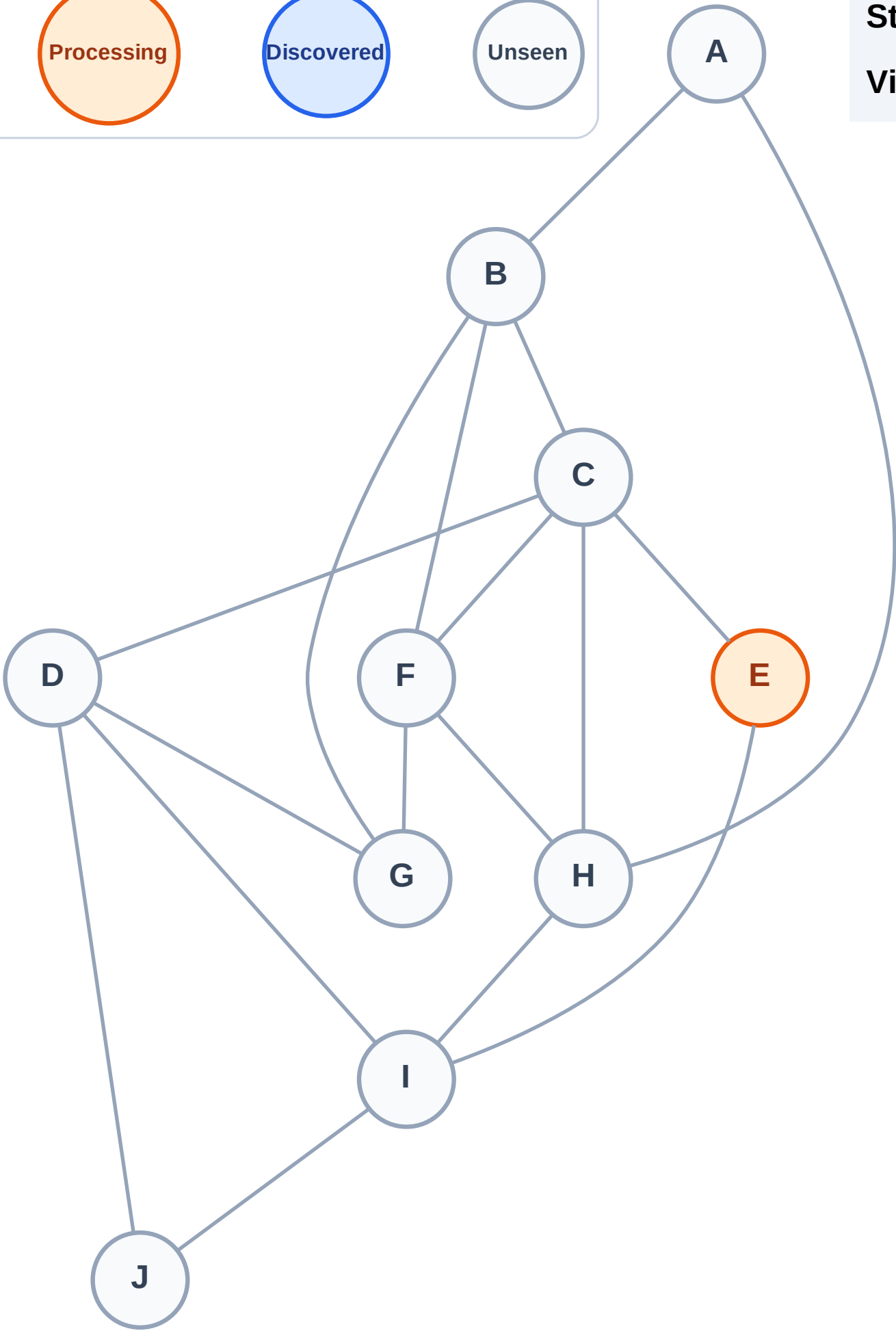
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover I, C from E

Legend

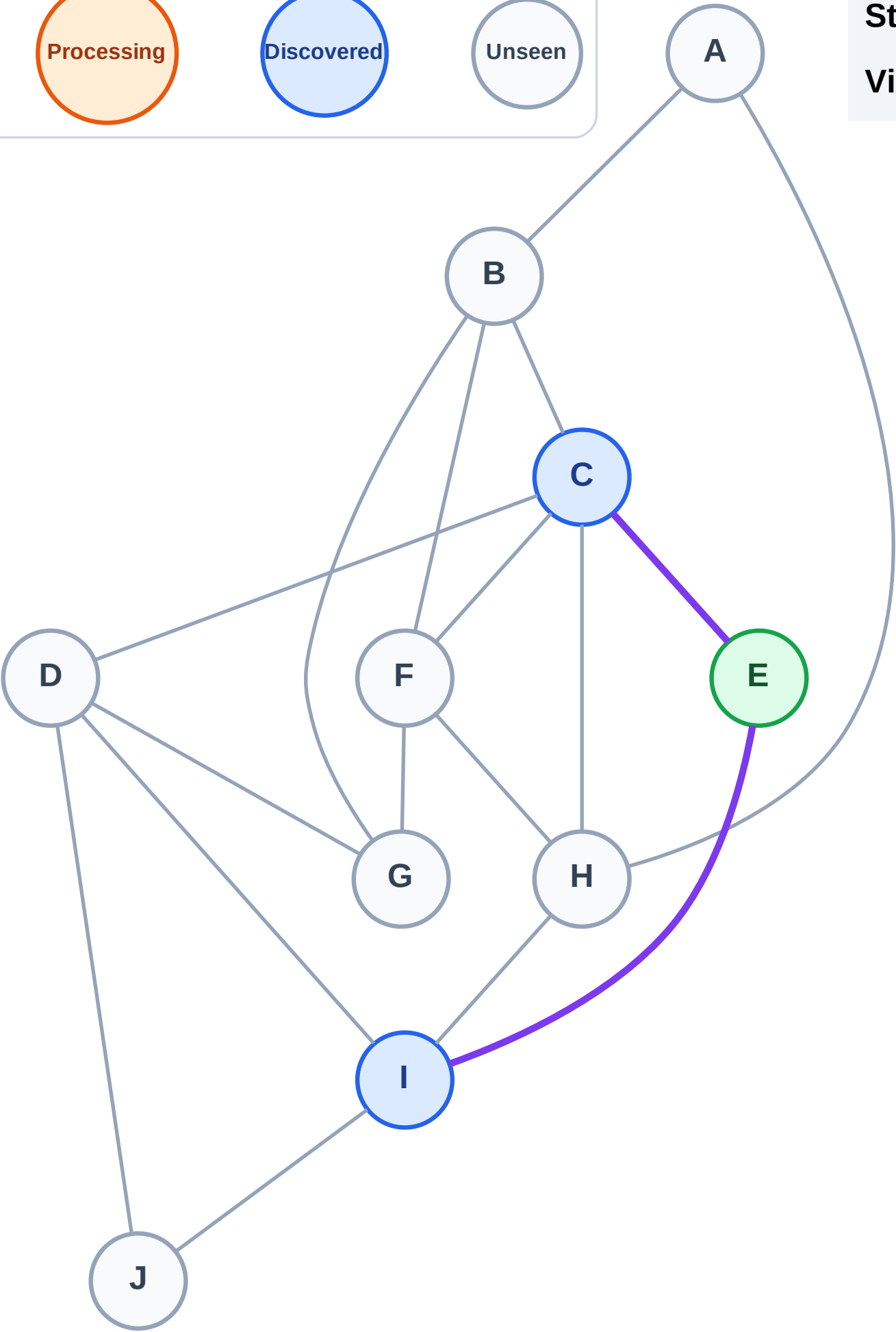
Complete

Processing

Discovered

Unseen

Stack (top at right): I | C
Visited: E



DFS Traversal

Step 4: Process node C

Legend

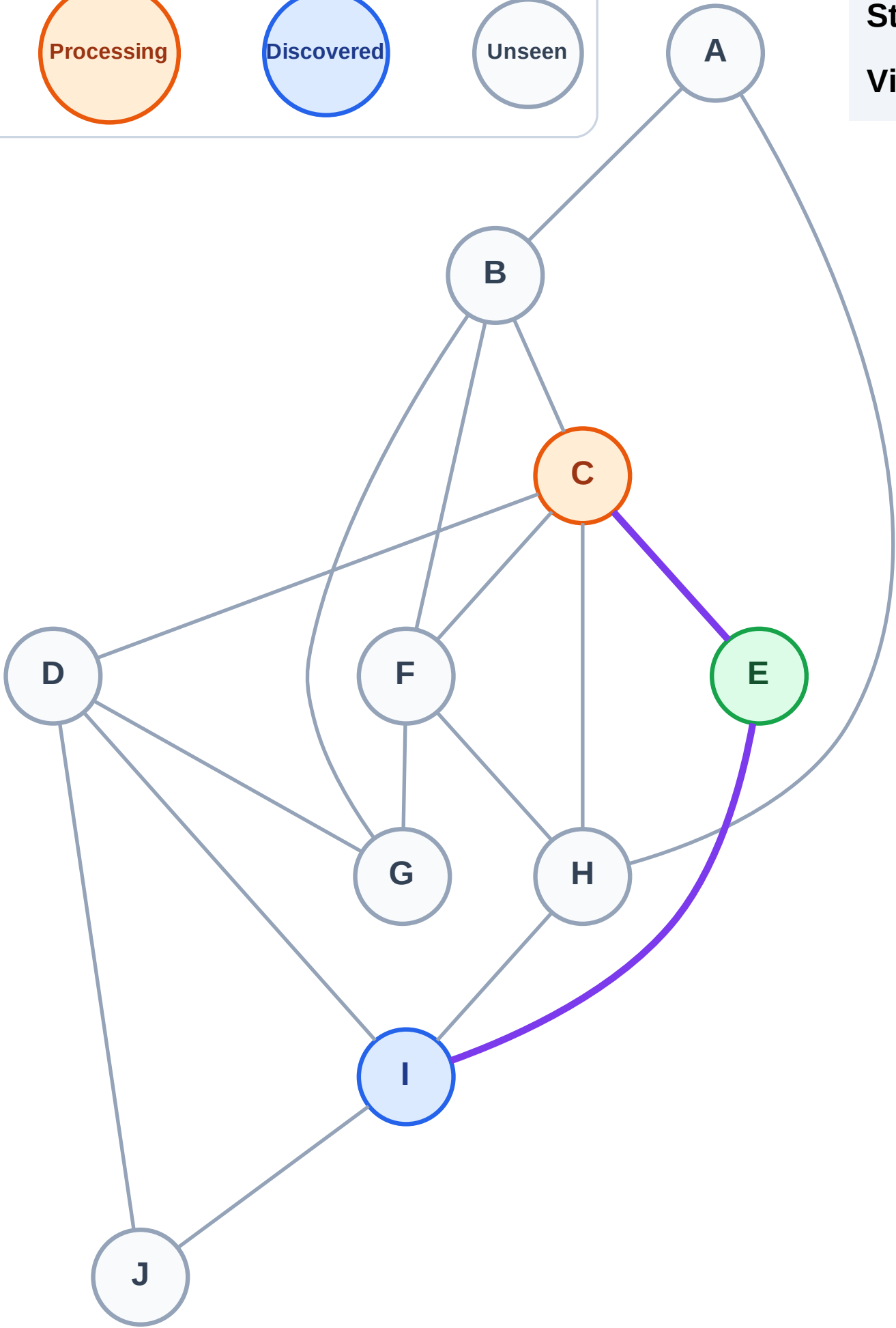
Complete

Processing

Discovered

Unseen

Stack (top at right): I
Visited: E



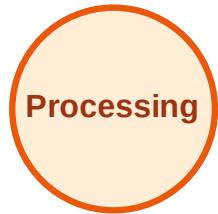
DFS Traversal

Step 5: Discover H, F, D, B from C

Legend



Complete



Processing



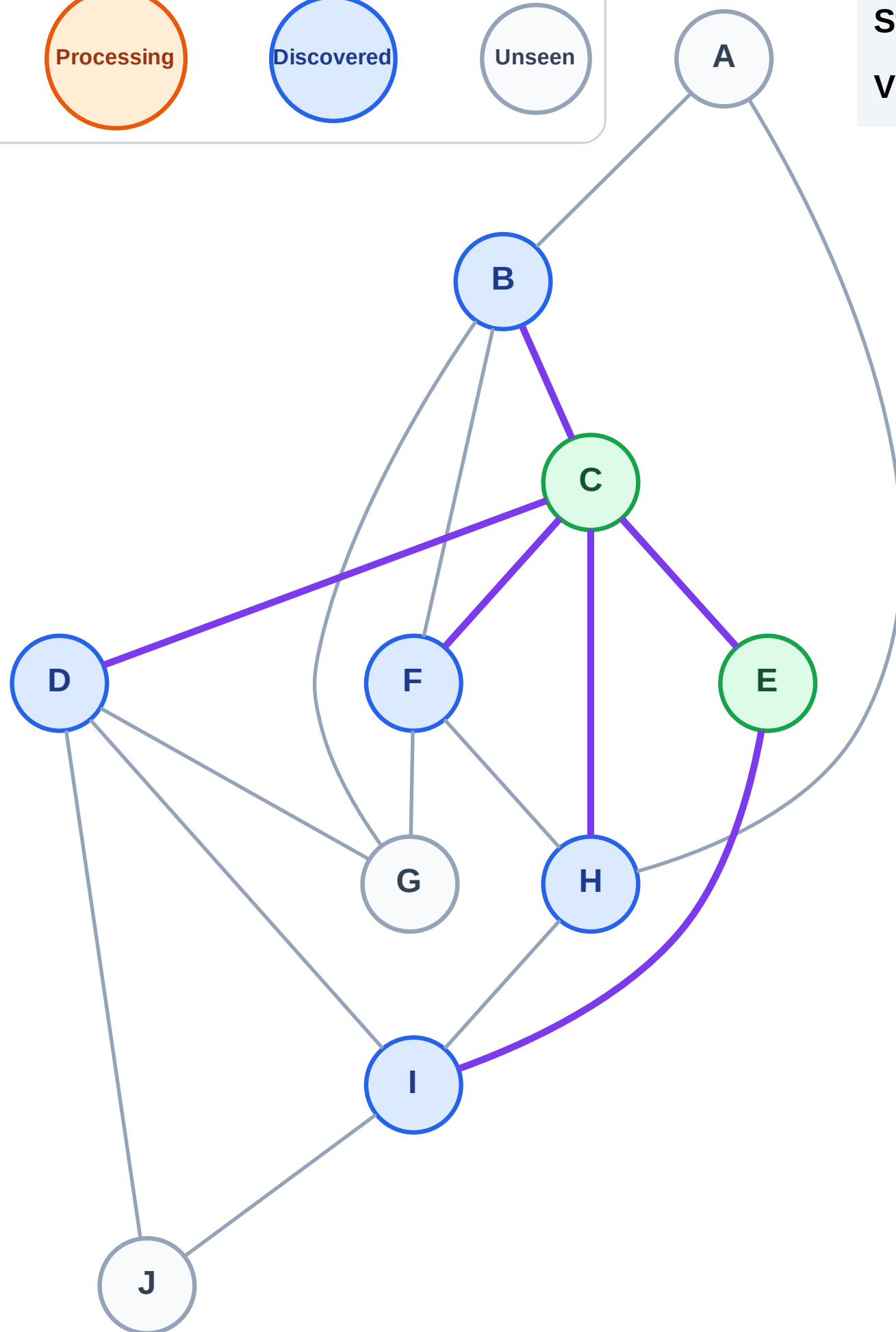
Discovered



Unseen

Stack (top at right): I | H | F | D | B

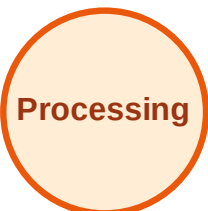
Visited: C, E



DFS Traversal

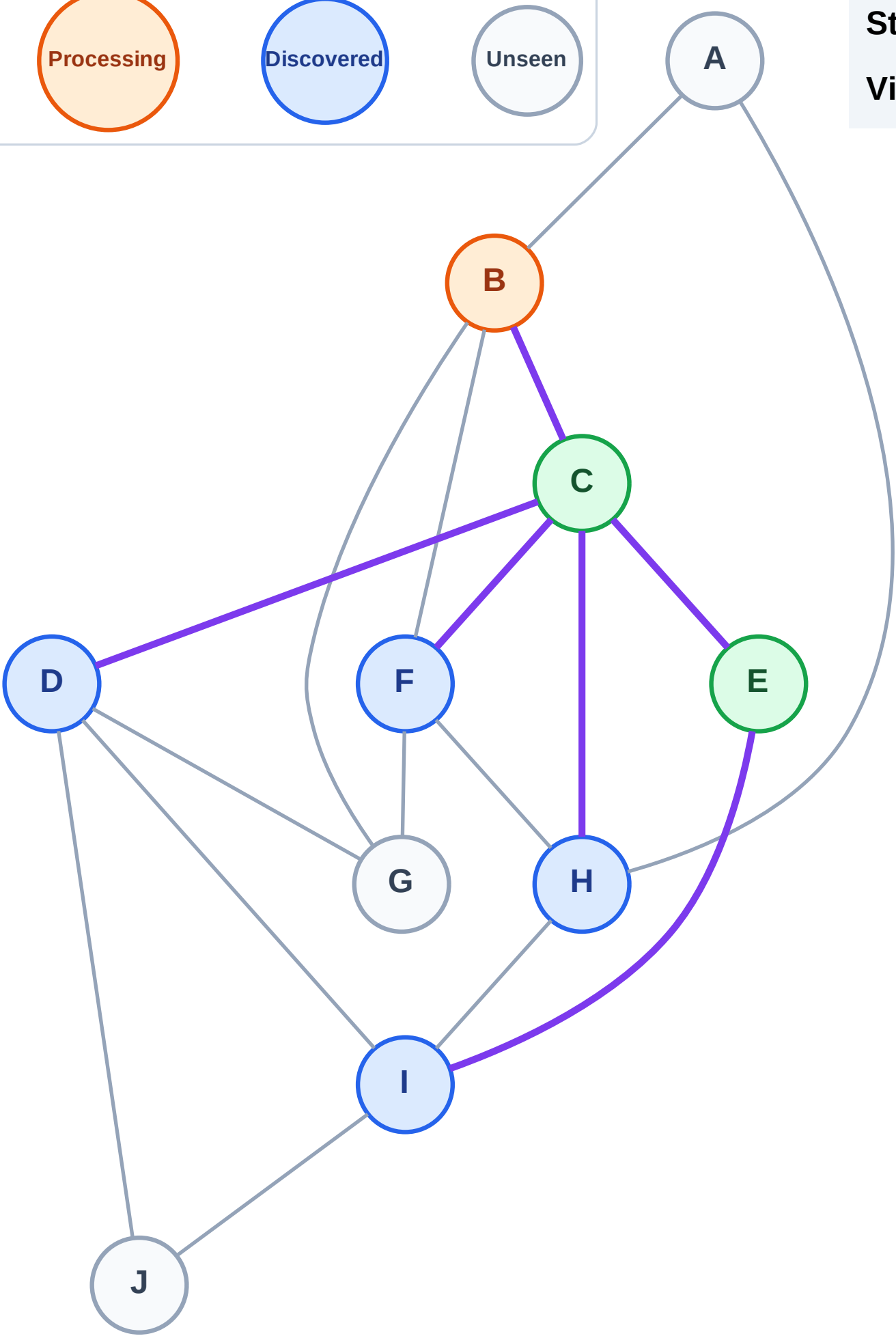
Step 6: Process node B

Legend



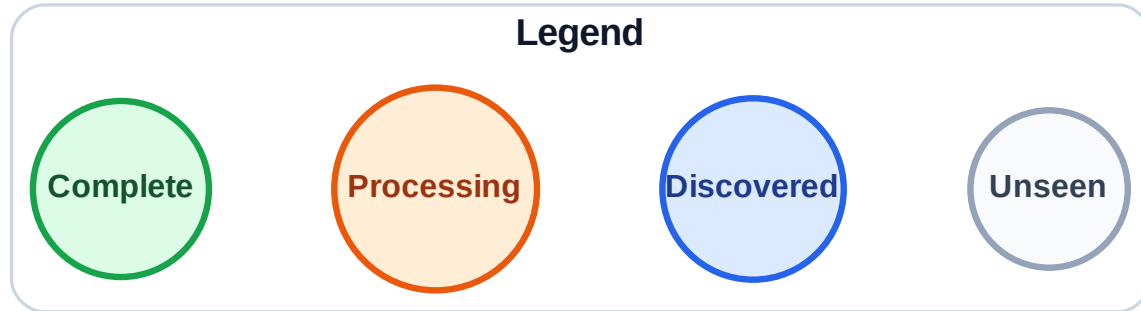
Stack (top at right): I | H | F | D

Visited: C, E



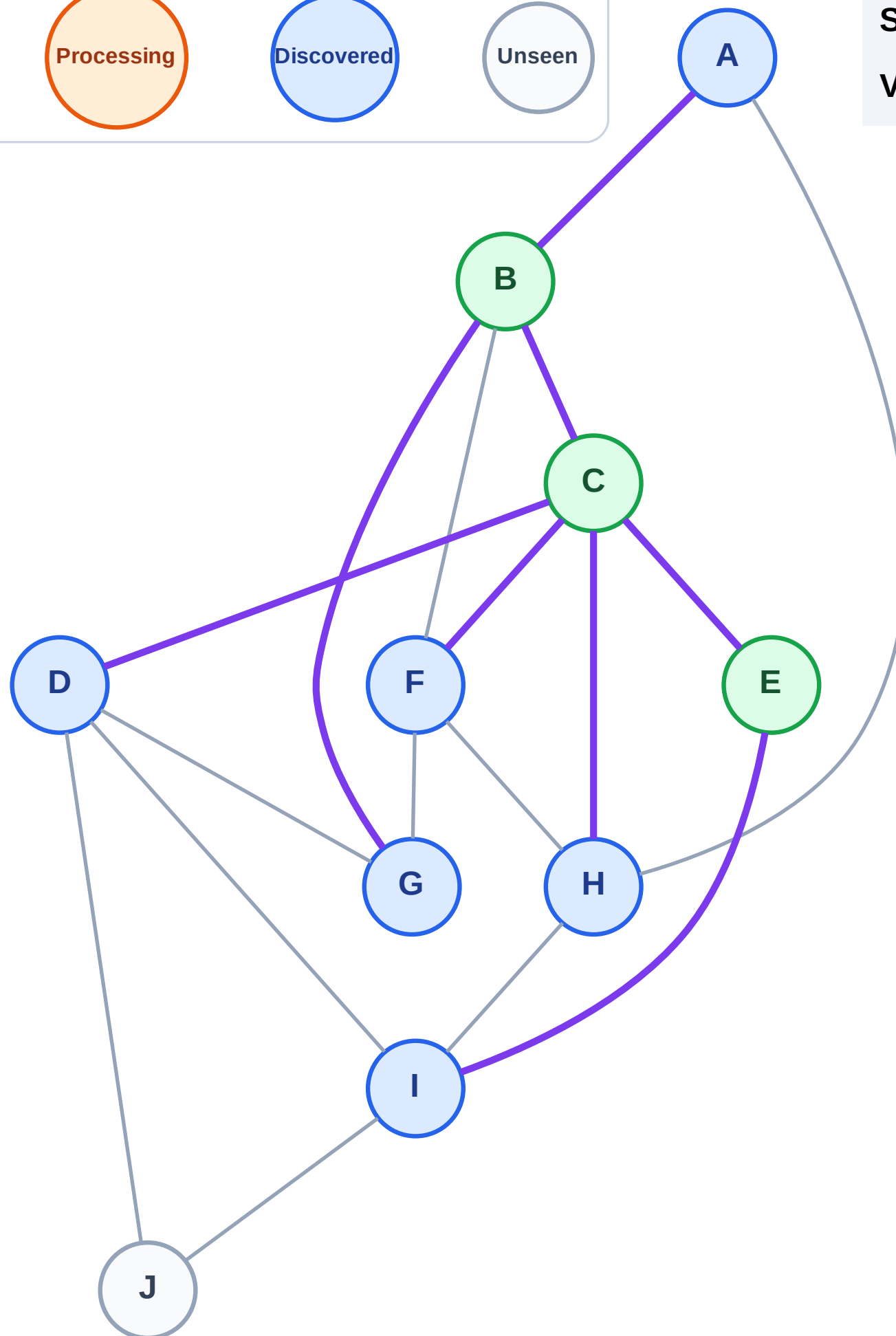
Step 7: Discover G, A from B

Step 7: Discover G, A from B



Stack (top at right): I | H | F | D | G | A

Visited: B, C, E



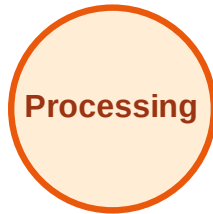
DFS Traversal

Step 8: Process node A

Legend



Complete



Processing



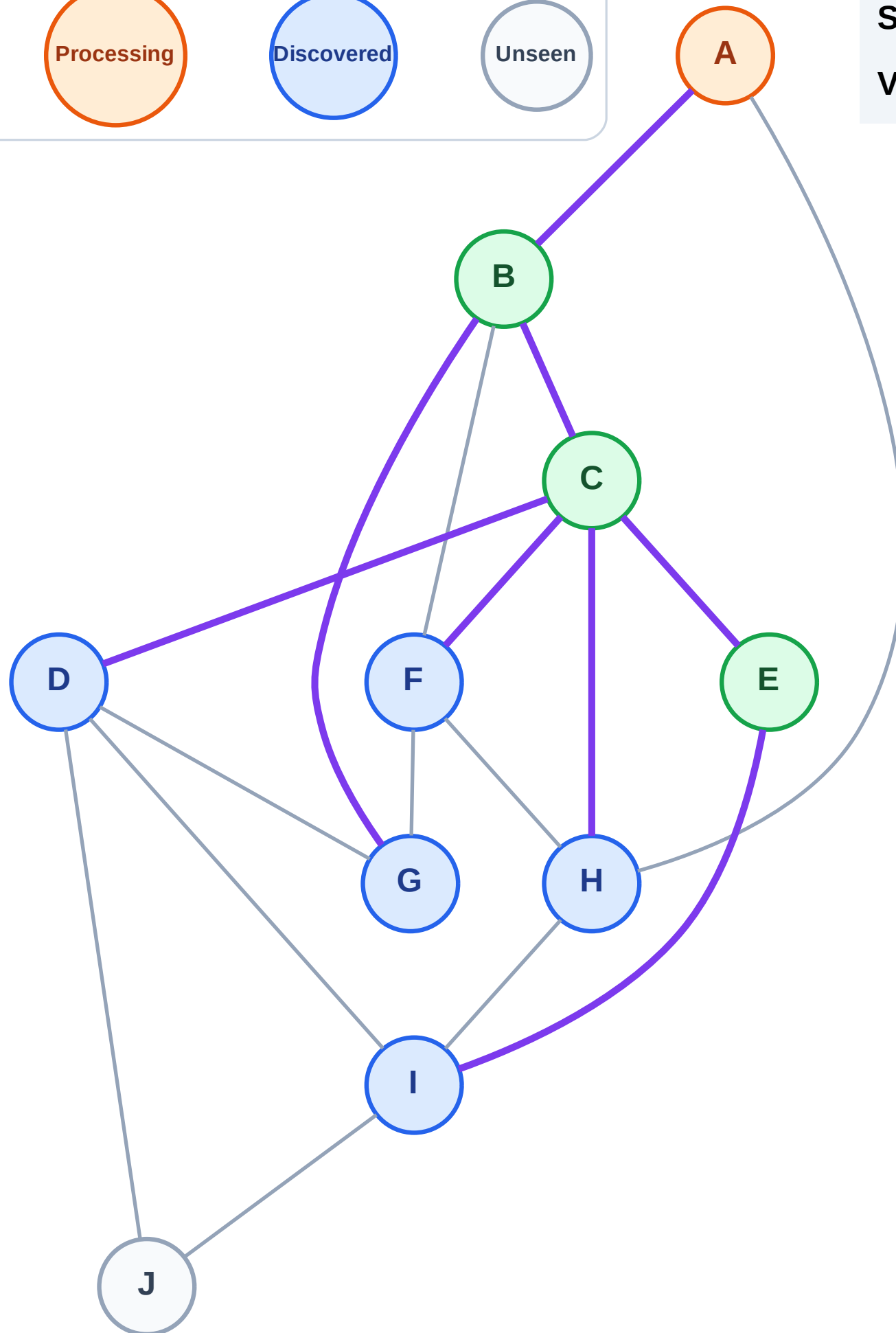
Discovered



Unseen

Stack (top at right): I | H | F | D | G

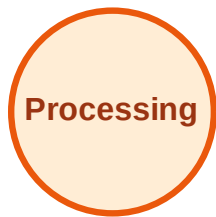
Visited: B, C, E



DFS Traversal

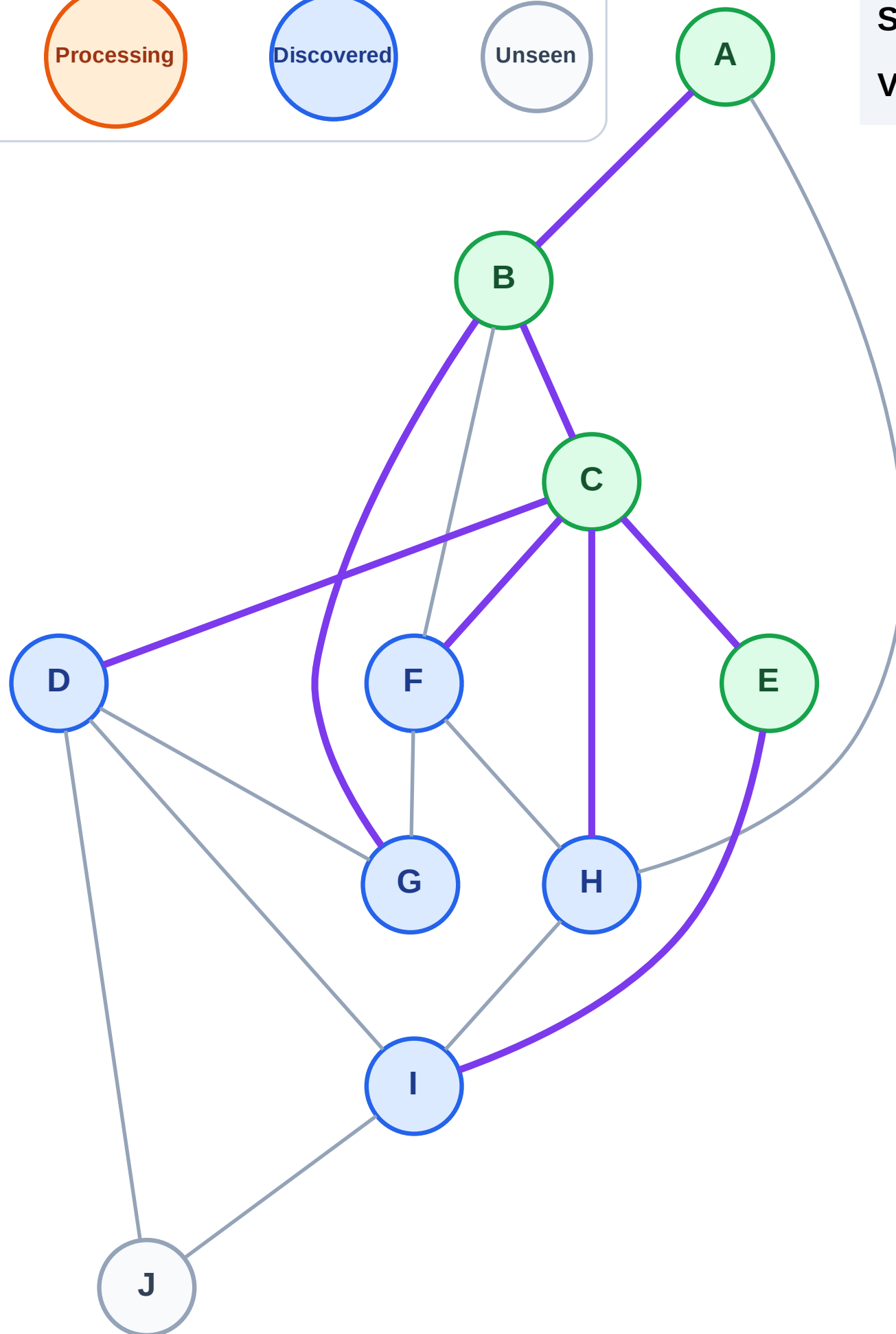
Step 9: No new neighbors from A

Legend



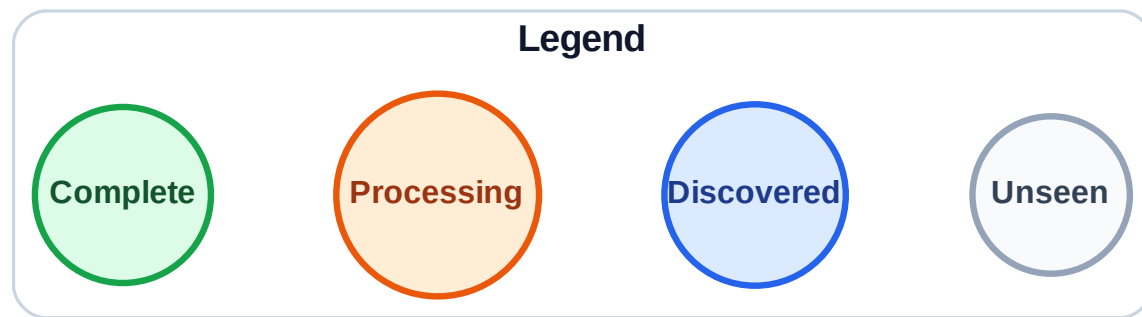
Stack (top at right): I | H | F | D | G

Visited: A, B, C, E

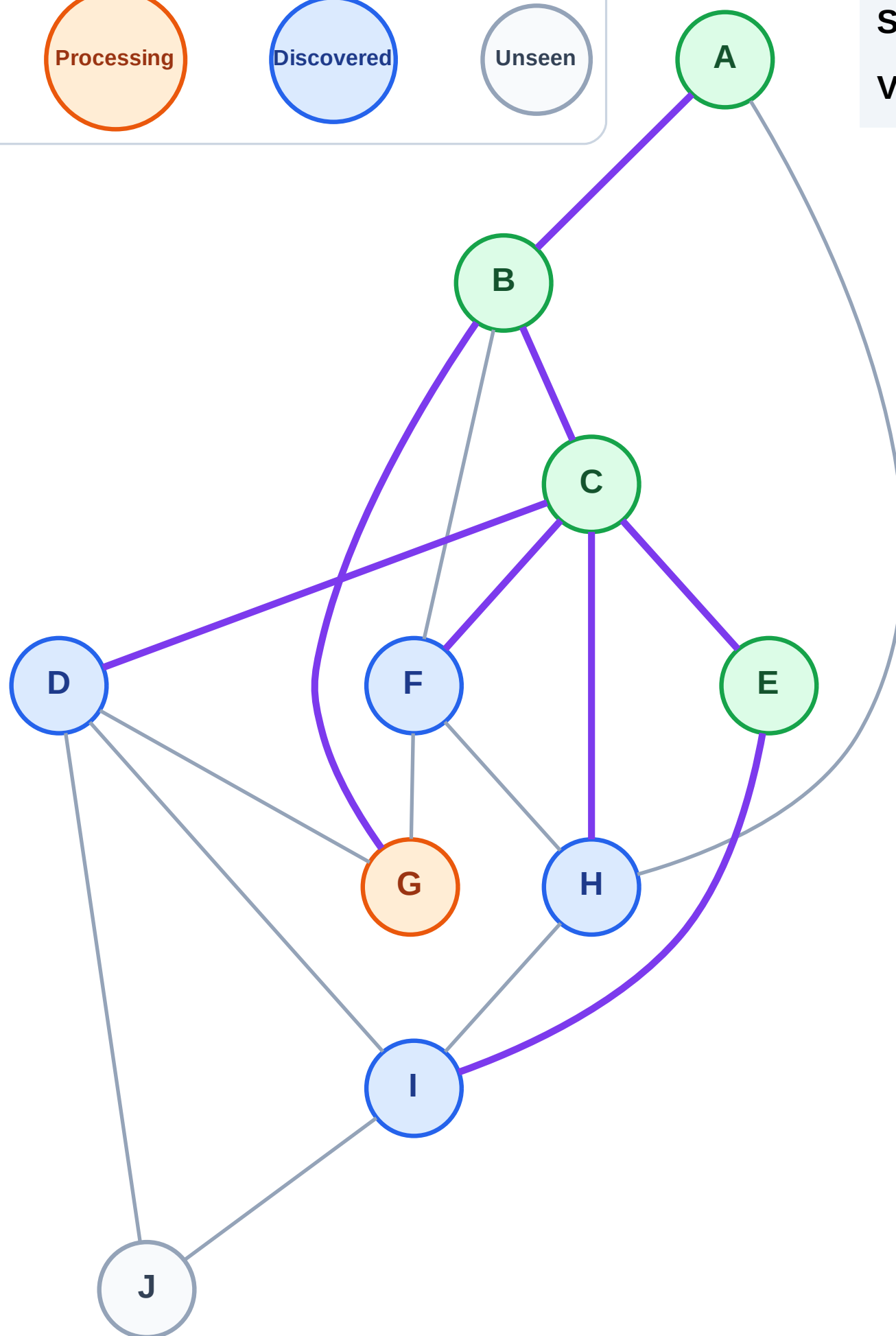


DFS Traversal

Step 10: Process node G



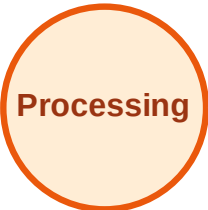
Stack (top at right): I | H | F | D
Visited: A, B, C, E



DFS Traversal

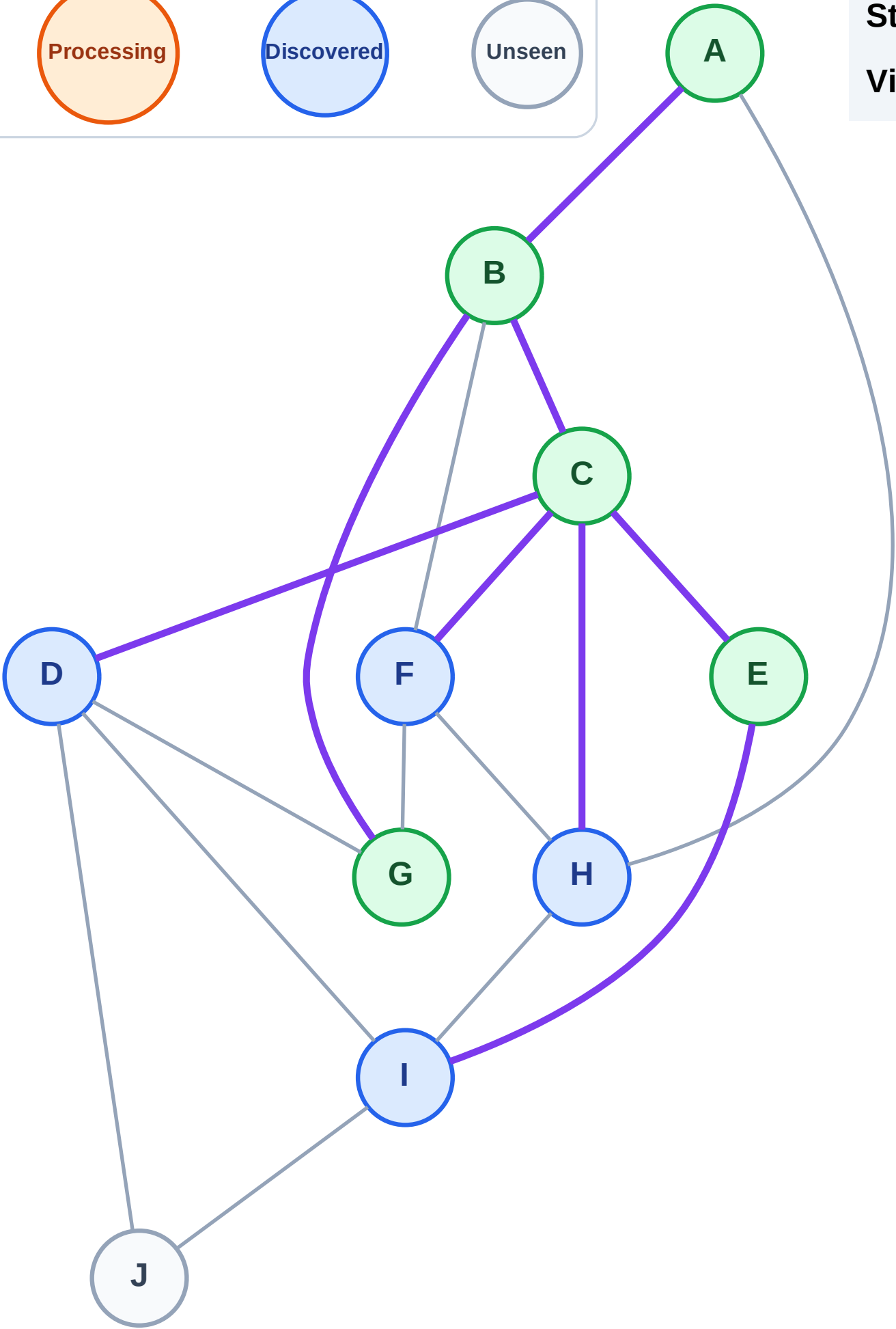
Step 11: No new neighbors from G

Legend



Stack (top at right): I | H | F | D

Visited: A, B, C, E, G



DFS Traversal

Step 12: Process node D

Legend

Complete

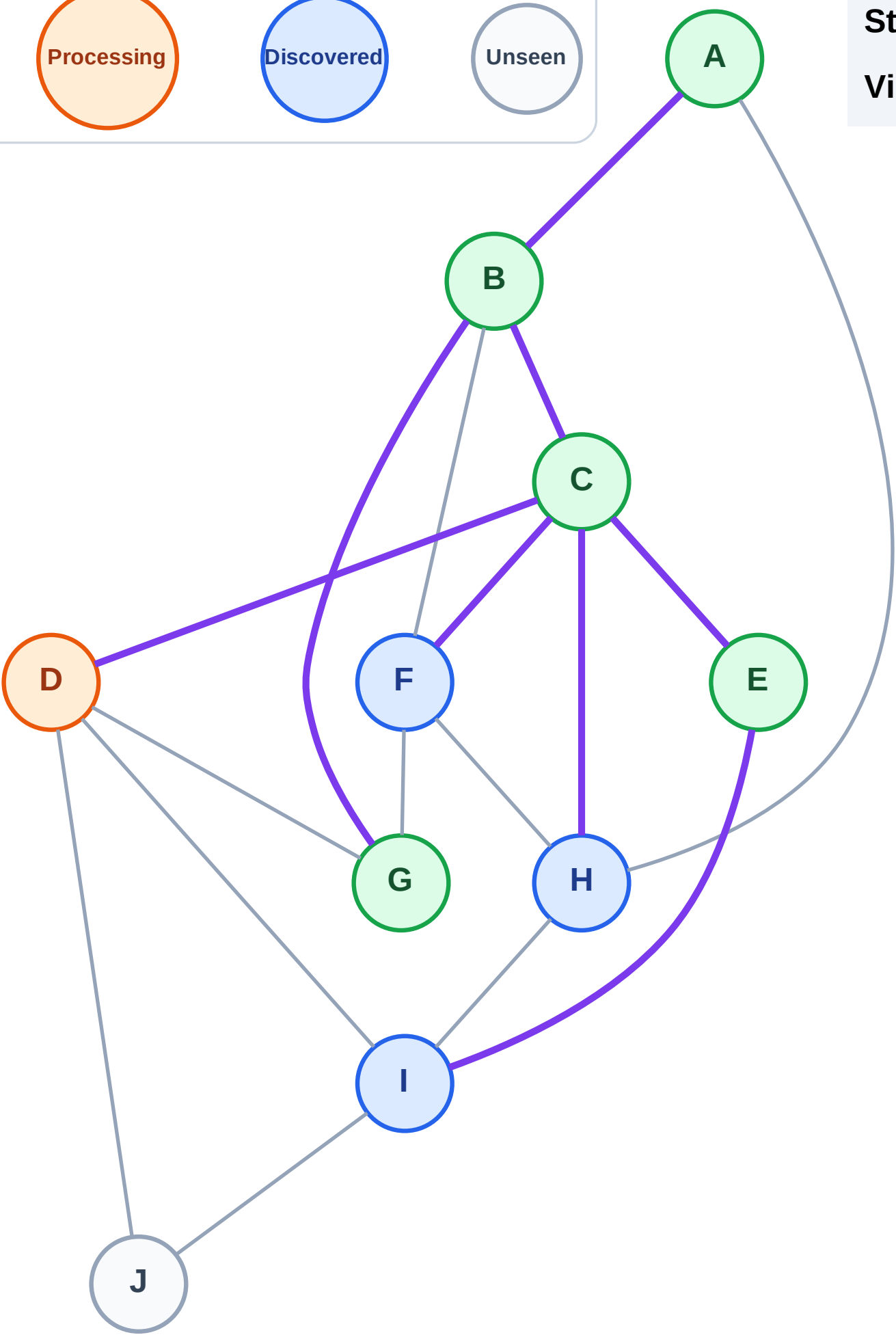
Processing

Discovered

Unseen

Stack (top at right): I | H | F

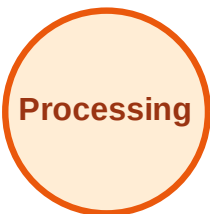
Visited: A, B, C, E, G



DFS Traversal

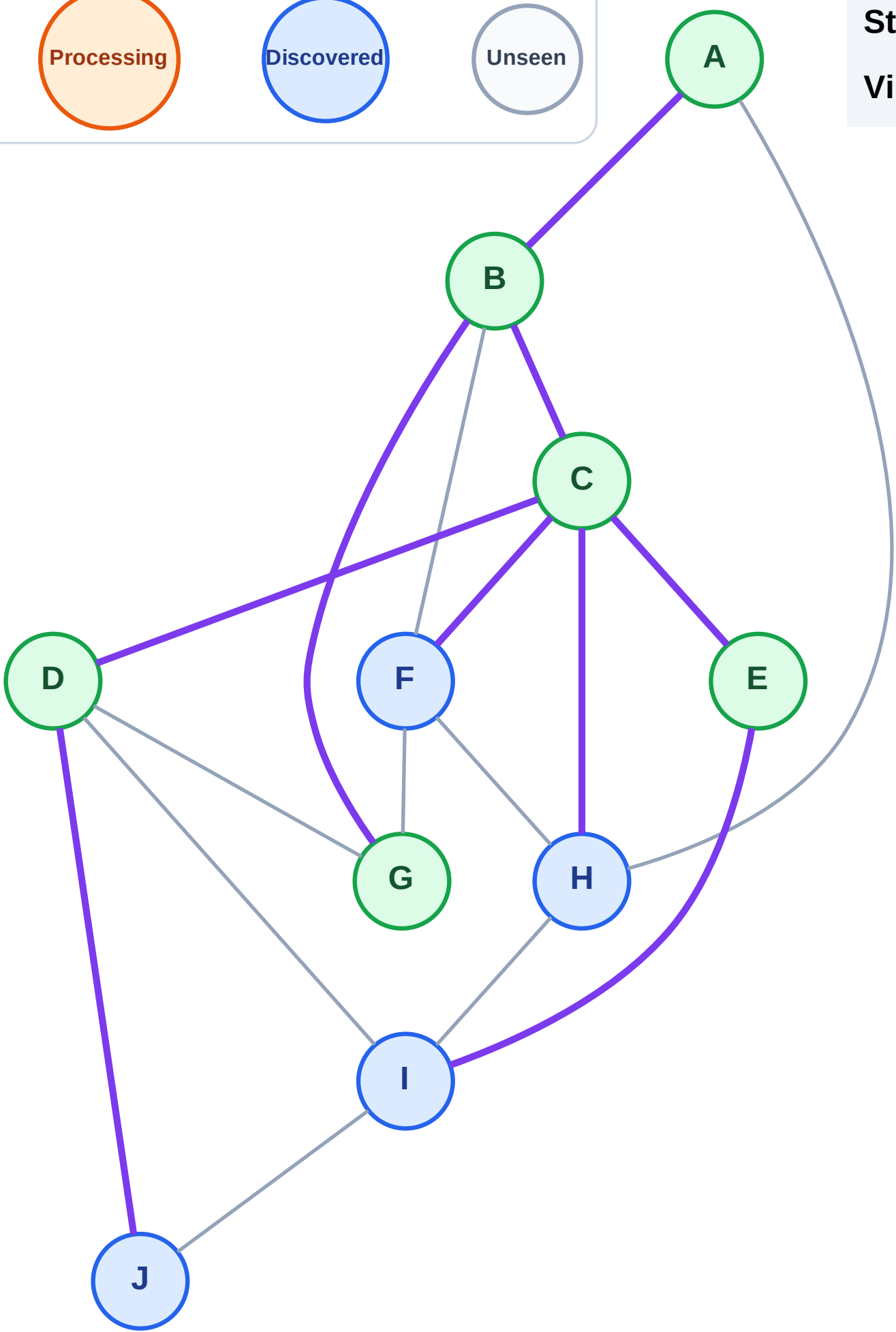
Step 13: Discover J from D

Legend



Stack (top at right): I | H | F | J

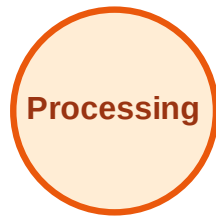
Visited: A, B, C, D, E, G



DFS Traversal

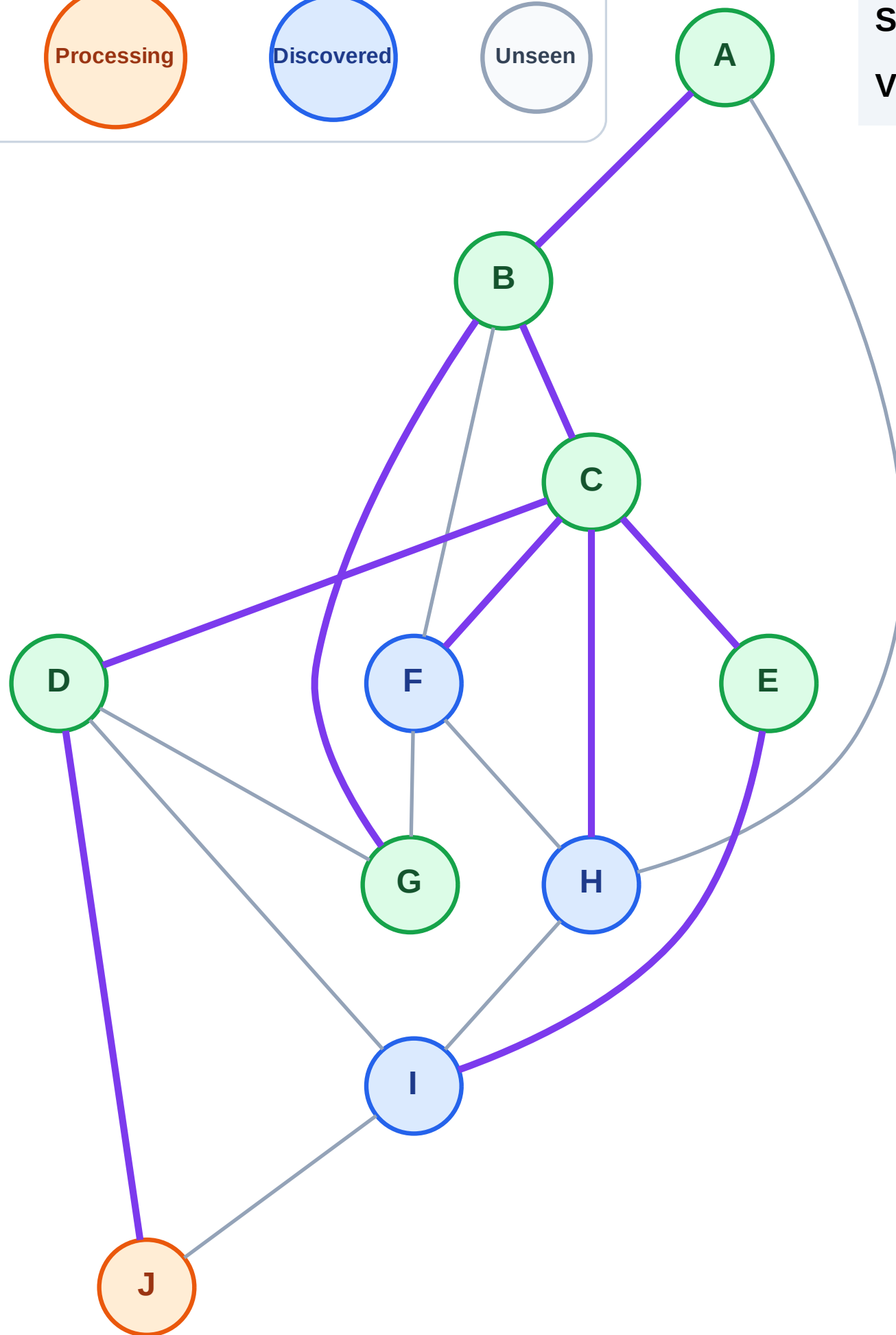
Step 14: Process node J

Legend



Stack (top at right): I | H | F

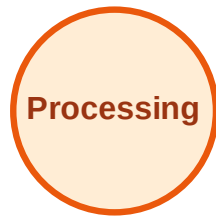
Visited: A, B, C, D, E, G



DFS Traversal

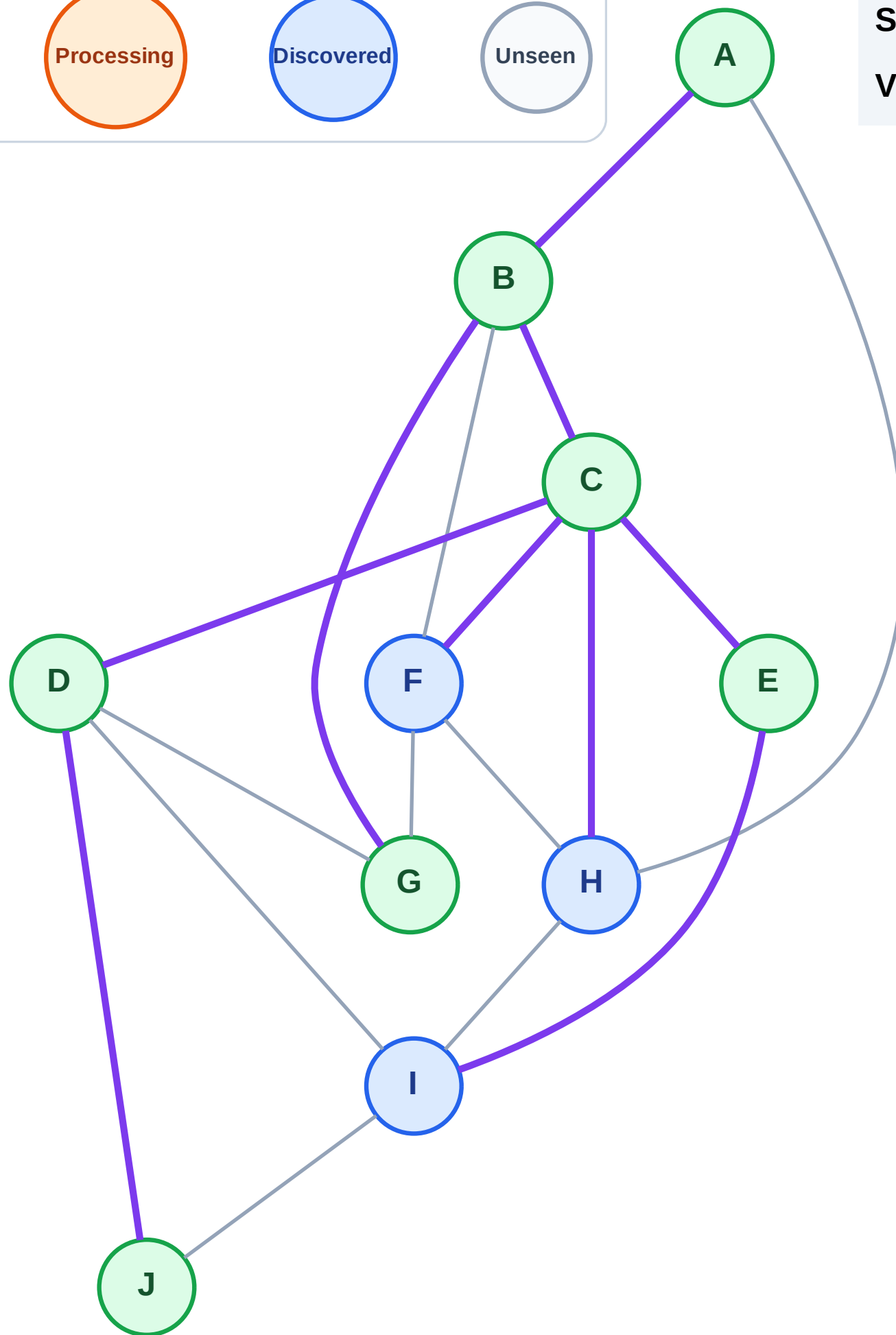
Step 15: No new neighbors from J

Legend



Stack (top at right): I | H | F

Visited: A, B, C, D, E, G, J



DFS Traversal

Step 16: Process node F

Legend

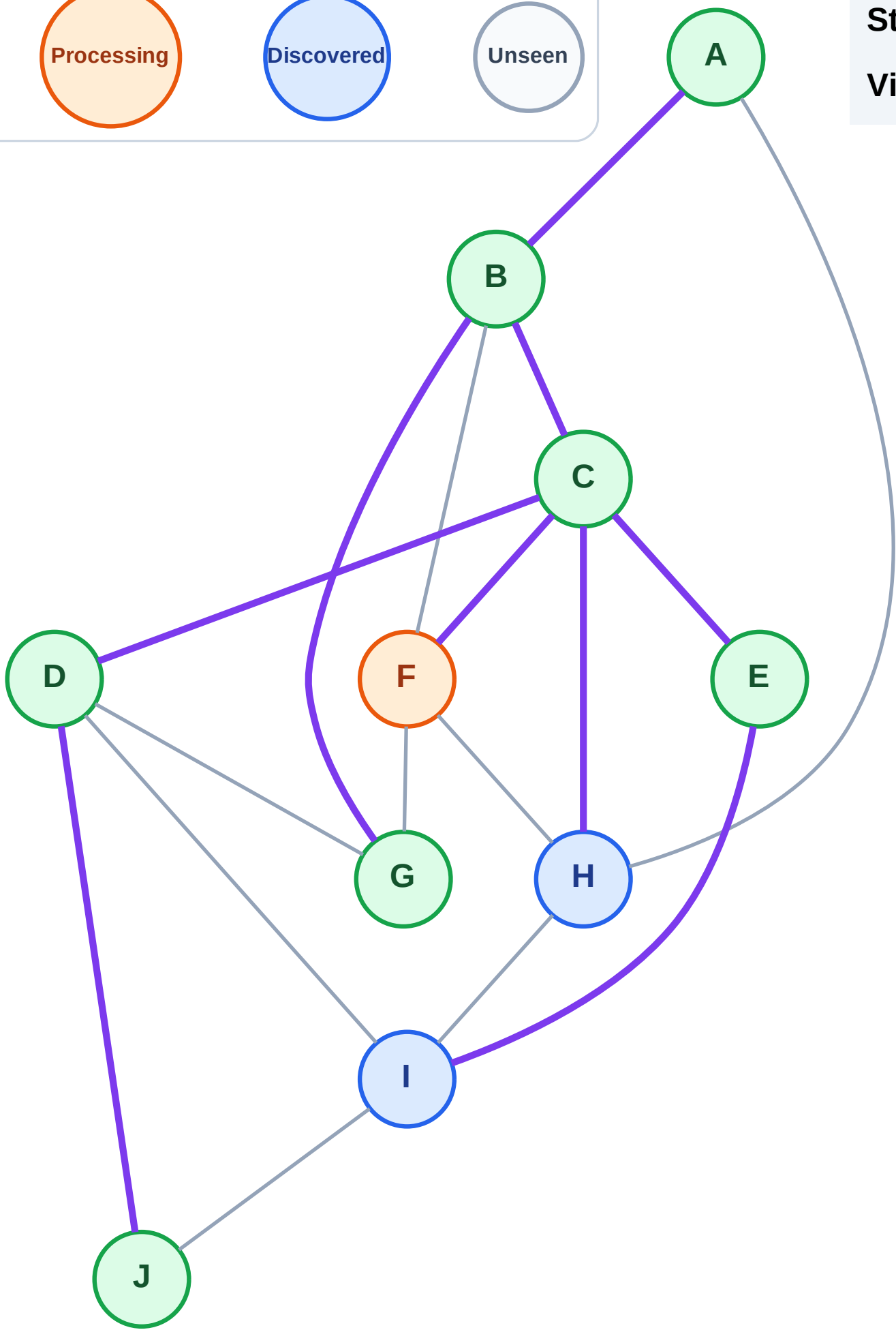
Complete

Processing

Discovered

Unseen

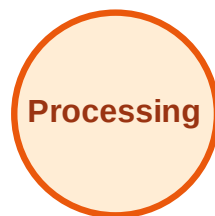
Stack (top at right): I | H
Visited: A, B, C, D, E, G, J



DFS Traversal

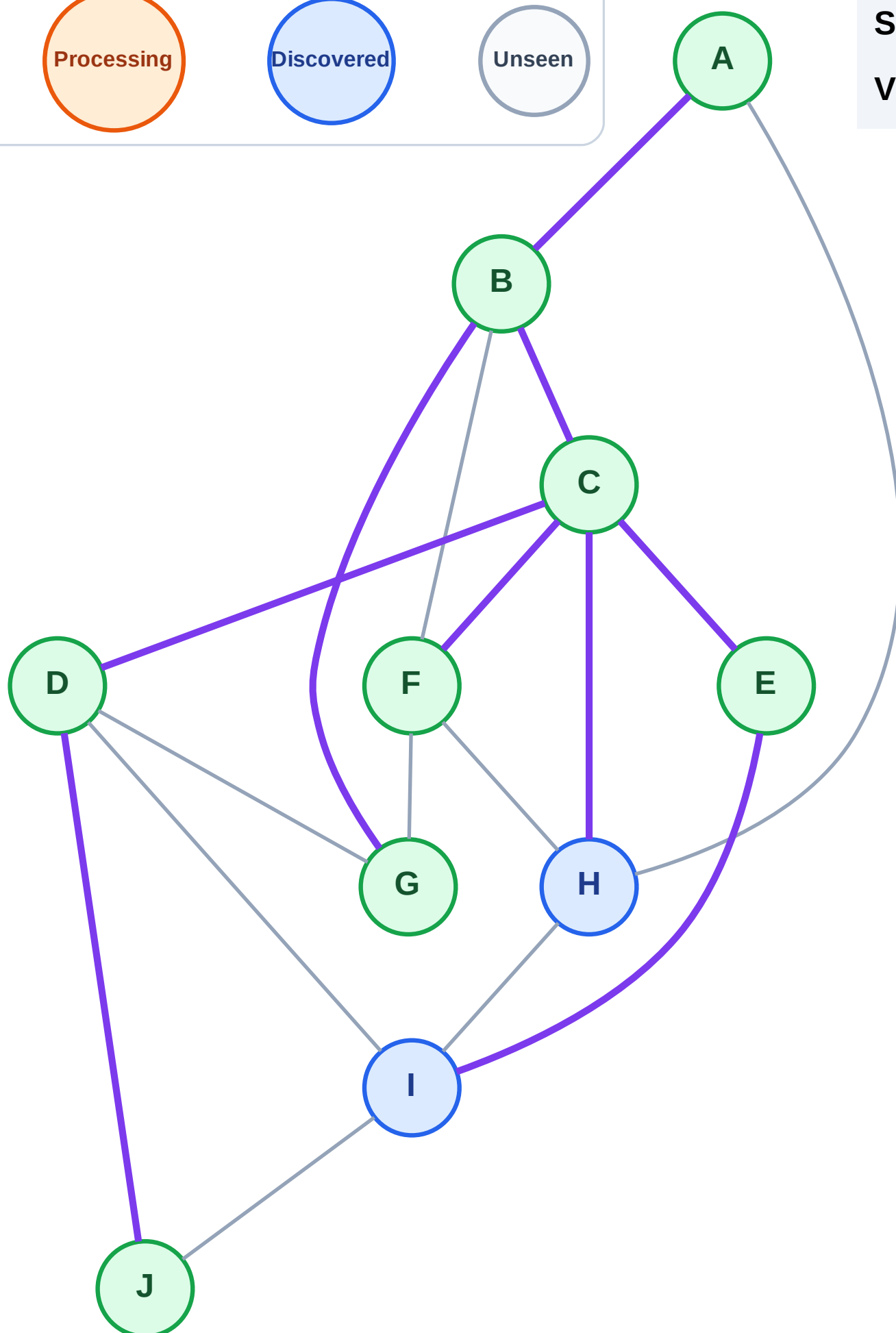
Step 17: No new neighbors from F

Legend



Stack (top at right): I | H

Visited: A, B, C, D, E, F, G, J



DFS Traversal

Step 18: Process node H

Legend

Complete

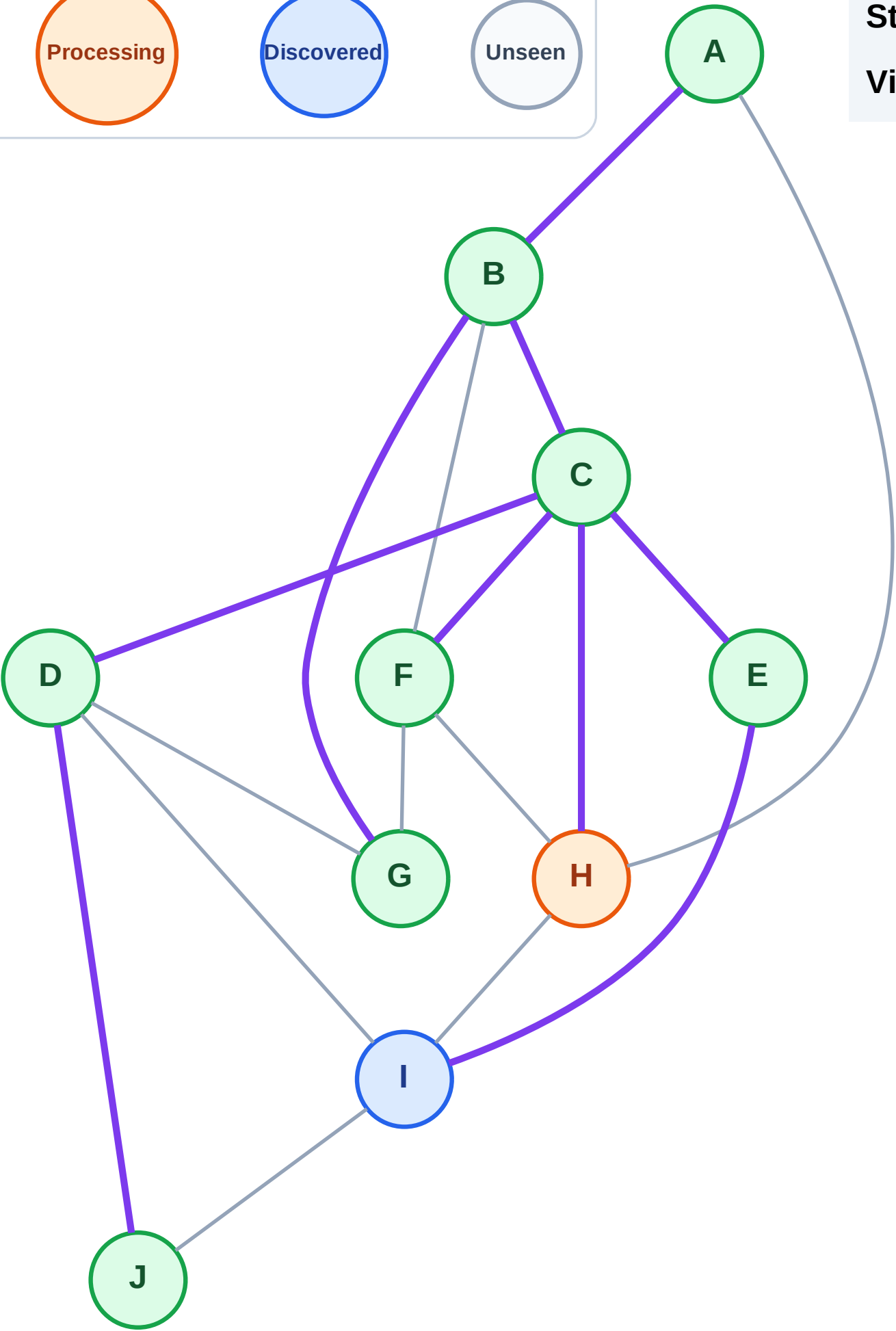
Processing

Discovered

Unseen

Stack (top at right): I

Visited: A, B, C, D, E, F, G, J

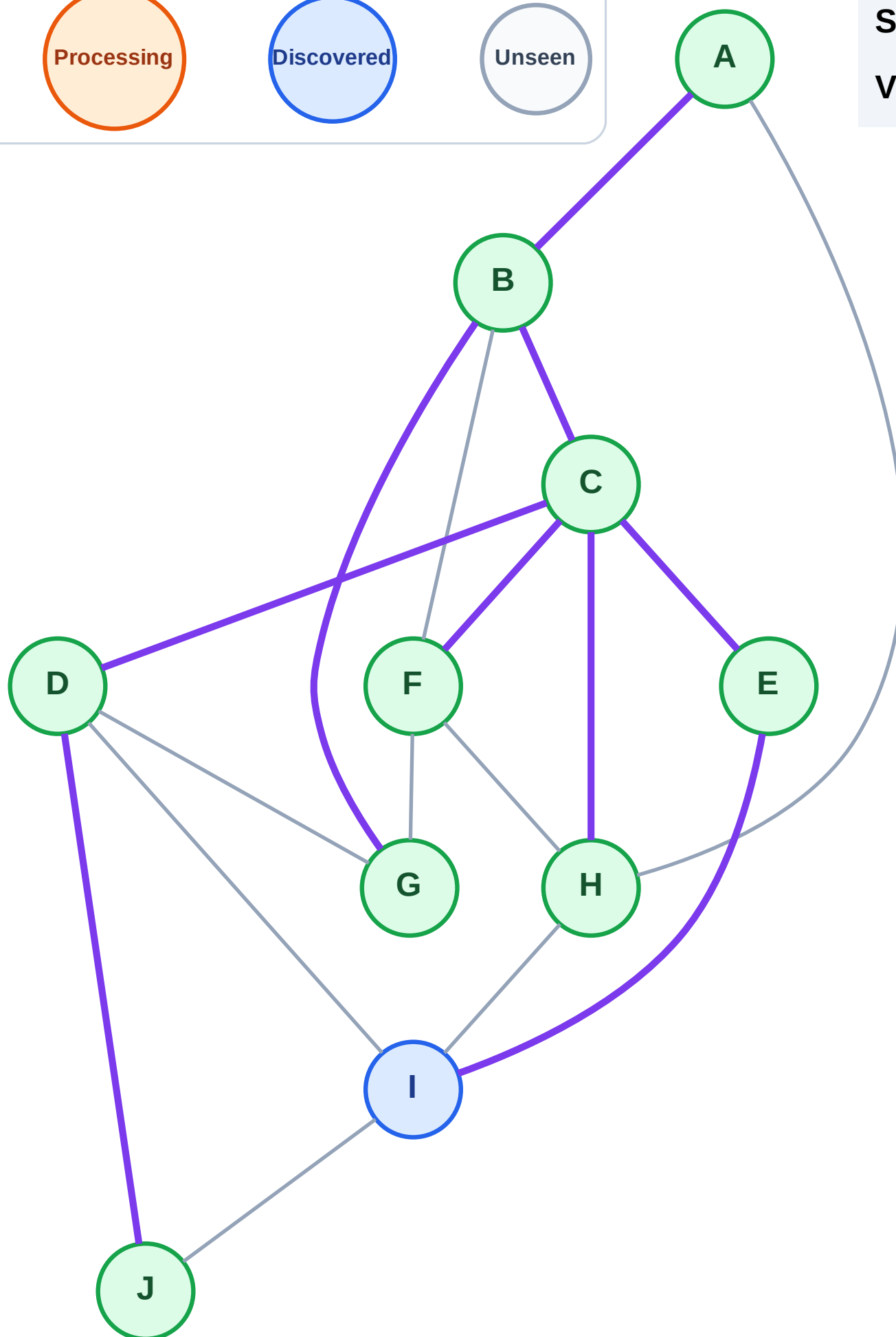


Step 19: No new neighbors from H

Legend



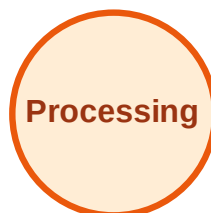
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

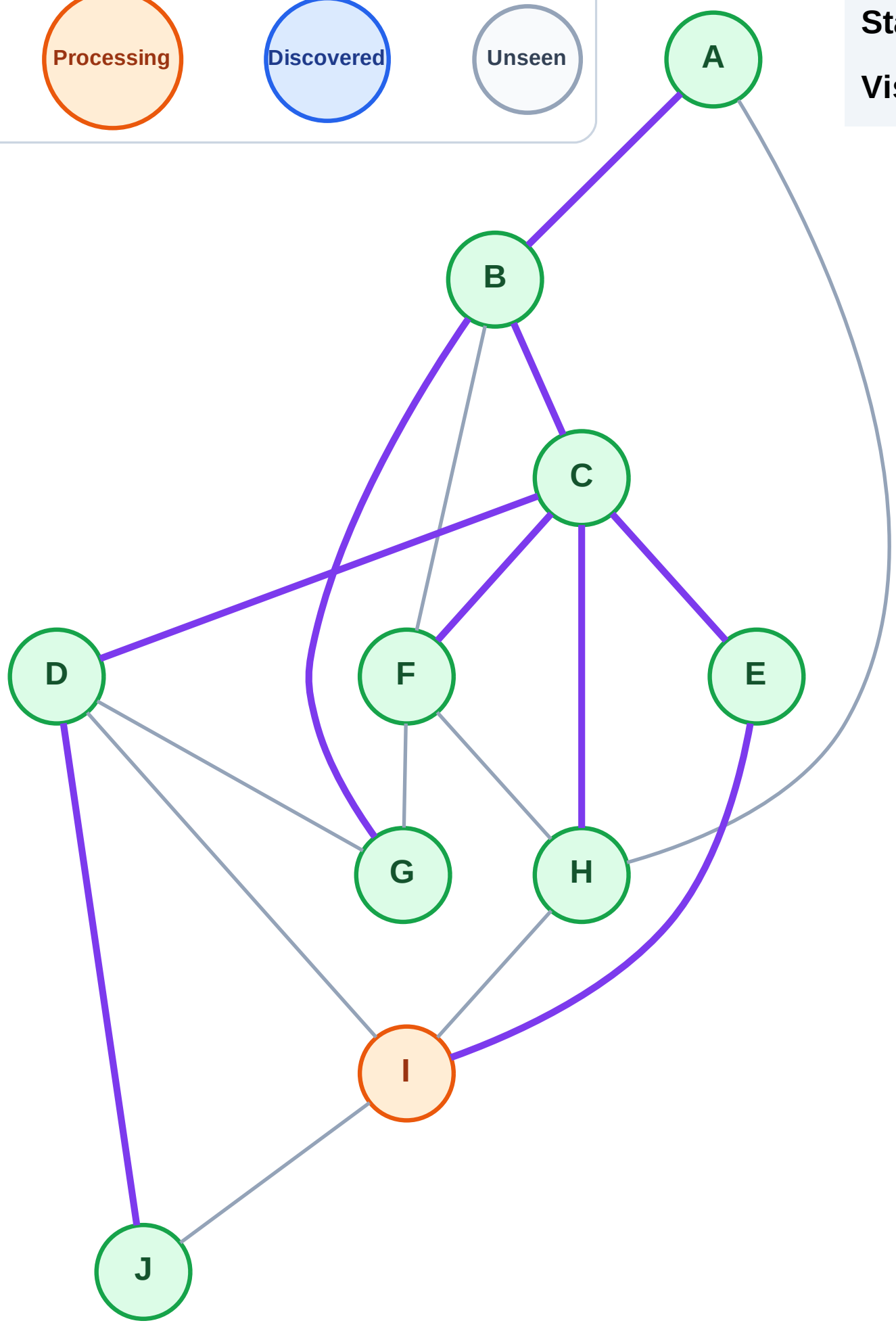
Step 20: Process node I

Legend



Stack (top at right): (empty)

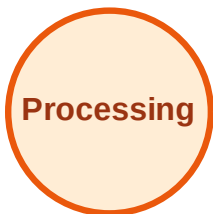
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

